Harmonic Map and PDE

University of Science and Technology of China

You Jiangcheng

Contents

Chapter 1 Crucial Techniques in Harmonic Map and PDE

1 Nash-Moser Iteration and Its Application to Regularity Theorem

In Han, Lin's book, "*Elliptic Partial Differential Equations*", a boundedness theorem can be written as

Theorem 1.1.1 (Local Boundedness Theorem)**.** *Suppose $a_{ij} \in L^\infty(B_1)$ and $c \in L^q(B_1)$ for some $q > \frac{n}{2}$ satisfy the following assumptions:*

$$a_{ij}(x)\xi_i\xi_j \geq \lambda|\xi|^2 \quad \text{for any } x \in B_1,\ \xi \in \mathbb{R}^n,$$

and

$$\|a_{ij}\|_{L^\infty} + \|c\|_{L^q} \leq \Lambda$$

for some positive constants λ and Λ. Suppose that $u \in H^1(B_1)$ is a subsolution in the following sense

$$\int_{B_1} a_{ij} D_i u D_j \varphi + c u \varphi \geq \int_{B_1} f \varphi \quad \text{for any } \varphi \in H_0^1(B_1) \text{ and } \varphi \geq 0 \text{ in } B_1.$$

If $f \in L^q(B_1)$, then $u^+ \in L^\infty_{\text{loc}}(B_1)$. Moreover, there holds for any $\theta \in (0,1)$ and any $p > 0$

$$\sup_{B_\theta} u^+ \leq C\left\{\frac{1}{(1-\theta)^{n/p}}\|u^+\|_{L^p(B_1)} + \|f\|_{L^q(B_1)}\right\}$$

where $C = C(n, \lambda, \Lambda, p, q)$ is a positive constant.

There are two approaches to prove this theorem, one by De Giorgi and the other by Moser. In Qing Jie's paper, "*a remark on the finite time singularity of the heat flow for harmonic maps*", a regularity theorem concerning heat flow for harmonic maps can be written as following, which can be proved via Nash-Moser iteration either:

Theorem 1.1.2. *Suppose that u is a solution to the heat flow for harmonic maps*

$$\frac{\partial}{\partial t}u = \Delta u + A(u)(\nabla u, \nabla u) \tag{1.1.2}$$

in P_{2r}, where $r \leq 1$. And suppose that $\sup_{P_{2r}}|\nabla u| \leq 2$. Then there exists a constant C s.t.

$$\sup_{P_r}|\nabla u|^2 \leq C\frac{1}{r^4}\int_{-4r^2}^{0}\int_{B_{2r}}|\nabla u|^2 dx \tag{1.1.3}$$

We denote the parabolic ball $\{(x,t) : -r^2 < t < 0; |x| < r\}$ by P_r.

Proof. Recall the Bochner identity for harmonic maps

$$\frac{\partial}{\partial t}\left(\frac{1}{2}|\nabla u|^2\right) = \Delta\left(\frac{1}{2}|\nabla u|^2\right) - |\nabla^2 u|^2 + Ric^M(\nabla u, \nabla u) + \left\langle R^N(\nabla u, \nabla u)\nabla u, \nabla u\right\rangle$$

We also have

$$\begin{aligned}\Delta\left(\frac{1}{2}|\nabla u|^2\right) &\leq |\nabla u|\Delta|\nabla u| + |\nabla\,|\nabla u||^2 \\ &= |\nabla u|\Delta|\nabla u| + \left|\nabla^2 u\frac{\nabla u}{|\nabla u|}\right|^2 \\ &\leq |\nabla u|\Delta|\nabla u| + \left|\nabla^2 u\right|^2\end{aligned}$$

Then we deduce

$$|\nabla u|\frac{\partial}{\partial t}|\nabla u| \leq |\nabla u|\Delta|\nabla u| + C|\nabla u|^2$$

In the above inequality, we use $\sup_{P_{2r}}|\nabla u| \leq 2$. If we denote $v = |\nabla u|$, then v satisfies

$$\frac{\partial}{\partial t}v \leq \Delta v + Cv \tag{1.1.4}$$

in P_{2r}. In the reminder of this proof, we set parameters $0 < r_0 < r_1 < r_2 < 2r$ and $0 < \tau_0 < \tau_2 < 4r^2$. Moreover, above inequality implies (put a cutoff function ϕ_1 satisfies

$\phi_1 = 1$ on $(-\tau_0, 0) \times B_{r_1}$ and $\operatorname{supp} \phi_1 \subset (-\tau_2, 0) \times B_{r_2}$):

$$\begin{aligned}
\frac{\partial}{\partial t} \int_{B_{r_2}} v^2 \phi_1^2 + \int_{B_{r_2}} |\nabla v|^2 \phi_1^2 &= \int_{B_{r_2}} 2 v v_t \phi_1^2 + \int_{B_{r_2}} 2 v^2 \phi_1 (\phi_1)_t + \int_{B_{r_2}} |\nabla v|^2 \phi_1^2 \\
&\leq \int_{B_{r_2}} 2 v \Delta v \phi_1^2 + \int_{B_{r_2}} 2 C v^2 \phi_1^2 + 2 v^2 \phi_1 |(\phi_1)_t| + |\nabla v|^2 \phi_1^2 \\
&= -\int_{B_{r_2}} 2 |\nabla v|^2 \phi_1^2 + 4 \phi_1 v \nabla \phi_1 \nabla v \\
&\quad + \int_{B_{r_2}} 2 C v^2 \phi_1^2 + 2 v^2 \phi_1 |(\phi_1)_t| + |\nabla v|^2 \phi_1^2 \\
&\leq \int_{B_{r_2}} 4 v^2 |\nabla \phi_1|^2 + \int_{B_{r_2}} 2 C v^2 \phi_1^2 + 2 v^2 \phi_1 |(\phi_1)_t| \\
&= 2C \int_{B_{r_2}} v^2 \phi_1^2 + \int_{B_{r_2}} 2 v^2 \left(2 |\nabla \phi_1|^2 + \phi_1 |(\phi_1)_t| \right)
\end{aligned}$$

Where we use $-\int 4\phi_1 v \nabla \phi_1 \nabla v \leq \int \phi_1^2 |\nabla v|^2 + 4 v^2 |\nabla \phi_1|^2$ in the second $\leq$. Integrate both sides on $[-\tau_2, 0]$, we obtain

$$\sup_{-\tau_0 \leq t \leq 0} \int_{B_{r_2}} v^2(\cdot, t) + \int_{-\tau_0}^0 \int_{B_{r_1}} |\nabla v|^2 \leq 2 \left(C + \frac{2}{(r_1 - r_2)^2} + \frac{1}{|\tau_0 - \tau_2|} \right) \int_{-\tau_2}^0 \int_{B_{r_2}} v^2 \tag{1.1.5}$$

Put a new cutoff function ϕ_2 which satisfies $\phi_2 = 1$ on $(-\tau_0, 0) \times B_{r_0}$ and $\operatorname{supp} \phi_2 \subset (-\tau_0, 0) \times B_{r_1}$, then Gagliardo-Nirenberg-Sobolev inequality reads

$$\begin{aligned}
\int_{B_{r_1}} v^4 \phi_2^4 &\leq C \left(\int_{B_{r_1}} \left| \nabla (v^2 \phi_2^2) \right| \right)^2 \\
&\leq C \left(\int_{B_{r_1}} 2 v |\nabla v| \phi_2^2 + 2 \phi_2 |\nabla \phi_2| v^2 \right)^2 \\
&\leq C \int_{B_{r_1}} v^2 \phi_2^2 \int_{B_{r_1}} \phi_2^2 |\nabla v|^2 + C \left(\int_{B_{r_1}} v^2 \phi_2 |\nabla \phi_2| \right)^2
\end{aligned}$$

Integrate both sides on $[-\tau_0, 0]$, we obtain

$$\int_{-\tau_0}^0 \int_{B_{r_0}} v^4 \leq C \left(\sup_{-\tau_0 \leq t \leq 0} \int_{B_{r_1}} v^2 \right) \left(\int_{-\tau_0}^0 \int_{B_{r_1}} |\nabla v|^2 + \frac{1}{(r_0 - r_1)^2} \int_{-\tau_0}^0 \int_{B_{r_1}} v^2 \right) \tag{1.1.6}$$

Combine (1.1.5) and (1.1.6) we obtain

$$\begin{aligned}
\int_{-\tau_0}^0 \int_{B_{r_0}} v^4 &\leq C \left(2 \left(C + \frac{2}{(r_1 - r_2)^2} + \frac{1}{|\tau_0 - \tau_2|} \right) \right) \\
&\quad \times \left(\frac{8 r^2}{(r_0 - r_1)^2} + 2 \right) \left(C + \frac{2}{(r_1 - r_2)^2} + \frac{1}{|\tau_0 - \tau_2|} \right) \left(\int_{-\tau_2}^0 \int_{B_{r_2}} v^2 \right)^2
\end{aligned} \tag{1.1.7}$$

We put $r_1 = 1/2(r_0 + r_2)$, then (1.1.7) can be written as

$$\int_{-\tau_0}^{0}\int_{B_{r_0}} v^4 \le C\left(2+\frac{32r^2}{(r_0-r_2)^2}\right)\left(C+\frac{8}{(r_0-r_2)^2}+\frac{1}{|\tau_0-\tau_2|}\right)^2\left(\int_{-\tau_2}^{0}\int_{B_{r_2}} v^2\right)^2 \tag{1.1.8}$$

Now we derive similar inequality as (1.1.4) by substituting v with v^k. Compute directly,

$$\begin{aligned}\frac{\partial}{\partial t}v^k = kv^{k-1}\frac{\partial}{\partial t}v &\le kv^{k-1}\Delta v + Ckv^k\\ &= \Delta v^k - k(k-1)v^{k-2}|\nabla v|^2 + Ckv^k\\ &\le \Delta v^k + Ckv^k\end{aligned}$$

We use $\Delta v^k = kv^{k-1}\Delta v + k(k-1)v^{k-2}|\nabla v|^2$. Since v^k satisfies (1.1.4), i.e. v^k is a subsolution, (1.1.8) implies

$$\begin{aligned}\left(\int_{-\tau_0}^{0}\int_{B_{r_0}} v^{4k}\right)^{\frac{1}{4k}} \le &\left(C\left(2+\frac{32r^2}{(r_0-r_2)^2}\right)\left(C+\frac{8}{(r_0-r_2)^2}+\frac{1}{|\tau_0-\tau_2|}\right)^2\right)^{\frac{1}{4k}}\\ &\times\left(\int_{-\tau_2}^{0}\int_{B_{r_2}} v^{2k}\right)^{\frac{1}{2k}}\end{aligned} \tag{1.1.9}$$

Now to carry out the Nash-Moser iteration, we choose parameters as following:

$$k = 2^l;\ r_0^l = r + \frac{r}{2^{l+1}};\ r_2^l = r + \frac{r}{2^l};\ \tau_0^l = (r_0^l)^2;\ \tau_2^l = (r_2^l)^2$$

Then (1.1.9) reads

$$\begin{aligned}\left(\int_{-\tau_0^l}^{0}\int_{B_{r_0^l}} v^{2^{l+2}}\right)^{\frac{1}{2^{l+2}}} &\le \left(C\left(2+2^{2l+7}\right)\left(C+\frac{2^{2l+5}}{r^2}+\frac{2^{2l+2}}{r^2(3+2^{l+2})}\right)^2\right)^{\frac{1}{2^{l+2}}}\\ &\quad\times\left(\int_{-\tau_2^l}^{0}\int_{B_{r_2^l}} v^{2^{l+1}}\right)^{\frac{1}{2^{l+1}}}\\ &\le C^{\frac{1}{2^{l+2}}}2^{\frac{2l+8}{2^{l+2}}+\frac{2l+6}{2^{l+1}}}\left(C+\frac{1}{r^2}\right)^{\frac{1}{2^{l+1}}}\left(\int_{-\tau_2^l}^{0}\int_{B_{r_2^l}} v^{2^{l+1}}\right)^{\frac{1}{2^{l+1}}}\end{aligned} \tag{1.1.10}$$

for $l = 0, 1, 2, \cdots$. Therefore

$$\begin{aligned}\left(\int_{-\tau_0^l}^{0}\int_{B_{r_0^l}} v^{2^{l+2}}\right)^{\frac{1}{2^{l+2}}} \le & C^{\sum_{p=0}^{l}\frac{1}{2^{p+2}}}2^{\sum_{p=0}^{l}\left(\frac{2p+8}{2^{p+2}}+\frac{2p+6}{2^{p+1}}\right)}\left(C+\frac{1}{r^2}\right)^{\sum_{p=0}^{l}\frac{1}{2^{p+1}}}\\ &\times\left(\int_{-4r^2}^{0}\int_{B_{2r}} v^2\right)^{\frac{1}{2}}\end{aligned} \tag{1.1.11}$$

Since the series above all converge, take $l \to +\infty$ we obtain

$$\sup_{P_r} v^2 \leq C_1 \left(C + \frac{1}{r^2} \right)^2 \int_{-4r^2}^{0} \int_{B_{2r}} v^2 \tag{1.1.12}$$

Then (1.1.3) follows from (1.1.12) immediately.

□

Remark 1. *We consider heat flow for harmonic maps which satisfies*

$$\frac{\partial}{\partial t} u = \Delta u + A(u)(\nabla u, \nabla u) \tag{1.1.13}$$

Actually, relevant results for linear parabolic equations still hold.

2 Oscillation Decay and Its Application to Hölder Continuity

It is well known that Let $\Omega \subset \mathbb{R}^n$ be a bounded domain, then for any $0 < \alpha < 1$, the embedding $C^\alpha(\Omega) \hookrightarrow L^\infty(\Omega)$ is compact. If we consider a more general space, such as a non-compact manifold or an unbounded Ω, this compact embedding may not hold — in such cases, decay conditions or weighted spaces must be considered.

Conversely, the embedding does not hold generally. However, When u is a solution to some elliptic equations, a specific Hölder norm of u on a smaller domain can be controlled by its L^∞ norm on a larger domain. This is an important regularity result for weak solutions $u \in H^1(B_1)$: **the Oscillation Decay Theorem**. This is one of the core results in the De Giorgi–Nash–Moser regularity theory, used to prove that weak solutions are Hölder continuous.

In this section, we study

$$\mathcal{L}u := \operatorname{div}(A(x)\nabla u(x)), \quad \forall\, x \in B_1. \tag{1.2.1}$$

for A uniformly elliptic, i.e., there exists $\lambda, \Lambda > 0$ s.t.

$$\lambda I \leq |A(x)| \leq \Lambda I \quad \forall\, x \in B_1$$

Most importantly, we assume no regularity on A, i.e. we only need $A \in L^\infty(B_1)$.

Throughout this section, we first derive a weak Harnack inequality (1.2.3), and use it to prove our main theorem: Oscillation Decay for solutions of elliptic equation (1.2.1). As a corollary, we will further demonstrate that weak solutions exhibit Hölder continuity, which will be stated by inequality (1.2.12).

Theorem 1.2.1 (Weak Harnack for Subsolution). *Assume $u \in H^1(B_1)$ a subsolution, i.e. $\mathcal{L}u \geq 0$ in B_1 in the weak sense, and assume there exists $\delta > 0$ s.t.*

$$m\left(\{u \leq 0\} \cap B_{1/2}\right) \geq \delta m\left(B_{1/2}\right) \tag{1.2.2}$$

Then the positive measure attracts our solution downwards, i.e., there exists $C(n, \lambda, \Lambda, \delta) > 0$ s.t.

$$\left\|u^+\right\|_{L^\infty(B_{1/4})} \leq (1 - C(\delta)) \left\|u^+\right\|_{L^\infty(B_1)}, \tag{1.2.3}$$

where $u^+ = \max\{u, 0\}$.

Proof. Leverage linearity, W.L.O.G. we may assume $\|u^+\|_{L^\infty(B_1)} = 1$ and it suffices to show that there exists $C(\delta) > 0$ s.t. $\forall\, x \in B_{1/4}$,

$$u^+ \leq 1 - C(\delta)$$

a.e. holds.

Put $t_k := 1 - 1/2^k$ for $k \geq 0$ and define

$$v_k := \frac{(u - t_k)^+}{1 - t_k},$$

An observation is that $\|v_k\|_{L^\infty(B_1)} = 1$. To see this, on the one hand,

$$|v_k| \leq 1 \ a.e. \Leftarrow (u - t_k)^+ \leq \frac{1}{2^k} \ a.e. \Leftarrow u - t_k \leq \frac{1}{2^k} \ a.e. \Leftarrow u \leq 1 \ a.e.$$

On the other hand, consider $E_\varepsilon := \{x \in B_1 \mid |u(x)| > 1 - \varepsilon\}$ and $E'_\varepsilon := \{x \in B_1 \mid$

$|v_k(x)| > 1-\varepsilon\}$. $\|u^+\|_{L^\infty(B_1)} = 1$ reads $m(E_\varepsilon) > 0$ for any $\varepsilon > 0$, thus

$$\begin{aligned}
m\left(E'_\varepsilon\right) &= m\left(\{x \in B_1 \mid |v_k(x)| > 1-\varepsilon\}\right) \\
&= m\left(\left\{x \in B_1 \mid (u(x)-t_k)^+ > \frac{1-\varepsilon}{2^k}\right\}\right) \\
&= m\left(\left\{x \in B_1 \mid u(x)-t_k > \frac{1-\varepsilon}{2^k}\right\} \bigcap \{x \in B_1 \mid u(x) > t_k\}\right) \\
&= m\left(\left\{x \in B_1 \mid u(x)-t_k > \frac{1-\varepsilon}{2^k}\right\} \bigcap \{x \in B_1 \mid u(x) > t_k\}\right) \\
&= m\left(\left\{x \in B_1 \mid u(x) > 1-\frac{\varepsilon}{2^k}\right\}\right) \\
&> 0
\end{aligned}$$

for any $\varepsilon > 0$.

Notice v_k remains a subsolution, so Local Boundedness Theorem (theorem 1.1.1) implies

$$\begin{aligned}
\|v_k\|_{L^\infty(B_{1/4})} &\leq C\|v_k\|_{L^2(B_{1/2})} \\
&\leq C\|v_k\|_{L^\infty(B_{1/2})} m\left(\{v_k > 0\} \cap B_{1/2}\right)^{\frac{1}{2}} \\
&\leq Cm\left(\{u > t_k\} \cap B_{1/2}\right)^{\frac{1}{2}}
\end{aligned}$$

Now we claim a fact, $\forall \varepsilon > 0$, there exists $k(\varepsilon) \gg 1$, such that

$$m\left(\{u > t_{k(\varepsilon)}\} \cap B_{1/2}\right) < \varepsilon \tag{1.2.4}$$

If so, pick $\varepsilon = \frac{1}{(2C)^2}$, then for $k = k(\varepsilon)$ as above we have $\forall$ *a.e.* $x \in B_{1/4}$,

$$\frac{(u-t_k)^+}{1-t_k} \leq Cm\left(\{u > t_k\} \cap B_{1/2}\right)^{\frac{1}{2}} < \frac{1}{2}$$

Hence

$$u \leq t_k + \frac{1}{2}(1-t_k) = 1 - \frac{1}{2^k} + \frac{1}{2^{k+1}} = 1 - \frac{1}{2^{k+1}}, \quad a.e.\ x \in B_{1/4}.$$

We pick $C(\delta) := 1/2^{k+1}$ to conclude. Furthermore, the dependence on δ shall be made clear once we prove the above fact.

Subsequently we prove above fact. For the sake of contradiction, assume there exists $\varepsilon > 0$ s.t. for any $k \geq 0$,

$$m\left(\{u > t_k\} \cap B_{1/2}\right) = m\left(\{v_k > 0\} \cap B_{1/2}\right) \geq \varepsilon \tag{1.2.5}$$

holds. We will derive the following

$$m\left(\left\{0 < v_k < \frac{1}{2}\right\} \bigcap B_{1/2}\right) \geq \beta \tag{1.2.6}$$

where $\beta > 0$ is independent of k, to reach a contradiction. To be more specific, notice that all sets $\{0 < v_k < 1/2\}$ are in fact disjoint

$$\left\{0 < v_k < \frac{1}{2}\right\} = \left\{0 < \frac{(u-t_k)^+}{1-t_k} < \frac{1}{2}\right\} = \{t_k < u < t_{k+1}\},$$

hence summing over yields a contradiction

$$\begin{aligned}
m\left(\{0<u<1\} \cap B_{1/2}\right) &= m\left(\bigcup_{k=0}^{\infty}\left(\left\{1-\frac{1}{2^k} < u < 1-\frac{1}{2^{k+1}}\right\} \bigcap B_{1/2}\right)\right) \\
&= m\left(\bigcup_{k=0}^{\infty}\left(\left\{0 < v_k < \frac{1}{2}\right\} \bigcap B_{1/2}\right)\right) \\
&= \sum_{k=0}^{\infty} m\left(\left\{0 < v_k < \frac{1}{2}\right\} \bigcap B_{1/2}\right) \\
&= \infty.
\end{aligned}$$

Our ultimate objective is to demonstrate inequality (1.2.6) rigorously.

We first give a lower bound for LHS of (1.2.6) with ∇v_k. By Caccioppoli Estimate,

$$\int_{B_{1/2}} |\nabla v_k|^2 \leq C \int_{B_1} v_k^2 \leq C \|v_k\|_{L^\infty(B_1)}^2 m\left(\{v_k > 0\} \cap B_1\right) \leq C$$

Above estimate holds uniformly in k. Therefore,

$$\begin{aligned}
\int_{\left\{0<v_k<\frac{1}{2}\right\} \cap B_{1/2}} |\nabla v_k| &\leq \left(\int_{B_{1/2}} |\nabla v_k|^2\right)^{\frac{1}{2}} m\left(\left\{0 < v_k < \frac{1}{2}\right\} \bigcap B_{1/2}\right)^{\frac{1}{2}} \\
&\leq C m\left(\left\{0 < v_k < \frac{1}{2}\right\} \bigcap B_{1/2}\right)^{\frac{1}{2}}.
\end{aligned}$$

Hence we take away

$$m\left(\left\{0 < v_k < \frac{1}{2}\right\} \bigcap B_{1/2}\right) \geq C\left(\int_{\left\{0<v_k<\frac{1}{2}\right\} \cap B_{1/2}} |\nabla v_k|\right)^2. \tag{1.2.7}$$

As an additional remark, we introduce the Caccioppoli Estimate.

Lemma 1.2.2. *Let $u \in H^1_{loc}(\Omega)$ be a weak solution to the homogeneous elliptic equation*

$$\operatorname{div}(A(x)\nabla u) = 0 \quad \textit{in } \Omega,$$

where $A(x)$ is uniformly elliptic and bounded. Then for any ball $B_{2R} \subset \Omega$, we have the following Caccioppoli Estimate:

$$\int_{B_R} |\nabla u|^2 \leq \frac{C}{R^2} \int_{B_{2R}} |u|^2,$$

where C depends only on the ellipticity constants and the dimension.

Next, we build a barrier function $\bar{v}_k$ as following:

$$\bar{v}_k \equiv 0 \ \{v_k \leq 0\}; \quad \bar{v}_k := v_k \ \{0 < v_k < \frac{1}{2}\}; \quad \bar{v}_k \equiv \frac{1}{2} \ \{v_k \geq \frac{1}{2}\}.$$

In this way we inherit the information from RHS of (1.2.7)

$$\begin{aligned}
\int_{\{0<v_k<\frac{1}{2}\}\cap B_{1/2}} |\nabla v_k| &= \int_{B_{1/2}} |\nabla \bar{v}_k| \\
&\geq C \int_{B_{1/2}} \left|\bar{v}_k - \fint_{B_{1/2}} \bar{v}_k\right| \\
&\geq C \int_{\{v_k \leq 0\}\cap B_{1/2}} \left|\fint_{B_{1/2}} \bar{v}_k\right| + C \int_{\{v_k \geq \frac{1}{2}\}\cap B_{1/2}} \left|\frac{1}{2} - \fint_{B_{1/2}} \bar{v}_k\right| \\
&\geq C \min\left\{ m\left(\{v_k \leq 0\} \cap B_{1/2}\right), m\left(\left\{v_k \geq \frac{1}{2}\right\} \bigcap B_{1/2}\right)\right\}. \quad (1.2.8)
\end{aligned}$$

Estimate the measure for the portion of $\{v_k \leq 0\}$ and $\{v_k \geq \frac{1}{2}\}$ in $B_{1/2}$ respectively via (1.2.2) and contradictory assumption (1.2.5):

$$\begin{aligned}
m\left(\{v_k \leq 0\} \cap B_{1/2}\right) &= m\left(\{(u - t_k)^+ \leq 0\} \cap B_{1/2}\right) \\
&= m\left(\{u \leq t_k\} \cap B_{1/2}\right) \\
&\geq m\left(\{u \leq 0\} \cap B_{1/2}\right) \\
&\geq \delta m\left(B_{1/2}\right)
\end{aligned}$$

$$\begin{aligned}
m\left(\left\{v_k \geq \frac{1}{2}\right\} \bigcap B_{1/2}\right) &= m\left(\left\{(u - t_k)^+ \geq \frac{1}{2^{k+1}}\right\} \bigcap B_{1/2}\right) \\
&\geq m\left(\left\{u \geq 1 - \frac{1}{2^k} + \frac{1}{2^{k+1}}\right\} \bigcap B_{1/2}\right) \\
&= m\left(\{u \geq t_{k+1}\} \cap B_{1/2}\right) \\
&= m\left(\{v_{k+1} \geq 0\} \cap B_{1/2}\right) \\
&\geq \varepsilon
\end{aligned}$$

Combine above estimate on measure for the portion of $\{v_k \leq 0\}$ and $\{v_k \geq \frac{1}{2}\}$ in $B_{1/2}$ with (1.2.7), (1.2.8), We have obtained a positive uniform lower bound of the measure for the focused set $\left\{0 < v_k < \frac{1}{2}\right\} \bigcap B_{1/2}$, thus (1.2.6) holds and we complete the proof. Notice the result depends on $\delta > 0$ in the sense that, if $\delta = 0$, our argument is inconclusive.

□

Based on the weak Harnack inequality derived earlier, we are now in a position to establish the oscillation decay for weak solutions of elliptic equation (1.2.1).

Theorem 1.2.3 (Oscillation Decay Theorem)**.** *Assume $u \in H^1(B_1)$ solves $\mathcal{L}u = 0$ in B_1 in the weak sense. Then for $\operatorname{osc}_{B_r} u := \sup_{B_r} u - \inf_{B_r} u$, there exists $\gamma = \gamma(n, \lambda, \Lambda) \in (0,1)$ s.t.*

$$\operatorname{osc}_{B_{1/4}} u \leq \gamma \operatorname{osc}_{B_{1/2}} u \tag{1.2.9}$$

Proof. In $B_{1/2}$, consider the midpoint value $\mu = \frac{1}{2}\left(\sup_{B_{1/2}} u + \inf_{B_{1/2}} u\right)$, by Lebesgue measure theory, almost every $x \in B_{1/2}$ satisfies $u(x) \geq \mu$ or $u(x) \leq \mu$. This classification essentially divides $B_{1/2}$ into two subsets with non-zero measure (unless u is constant), providing a basis for subsequent extremal analysis.

For $0 < r < 1$, denote

$$\alpha(r) := \sup_{x \in B_r} u, \quad \beta(r) := \inf_{x \in B_r} u,$$

then we have two corresponding relations for both $u(x) \geq \mu$ and $u(x) \leq \mu$. In the first case ($u(x) \geq \mu$)

$$\begin{aligned} u \geq \frac{1}{2}\left(\alpha\left(\frac{1}{2}\right) + \beta\left(\frac{1}{2}\right)\right) &\iff u - \alpha\left(\frac{1}{2}\right) \geq -\frac{1}{2}\left(\alpha\left(\frac{1}{2}\right) - \beta\left(\frac{1}{2}\right)\right) \\ &\iff \frac{\alpha\left(\frac{1}{2}\right) - u}{\alpha\left(\frac{1}{2}\right) - \beta\left(\frac{1}{2}\right)} \leq \frac{1}{2}. \end{aligned}$$

Similarly, in the second case ($u(x) \leq \mu$)

$$\begin{aligned} u \leq \frac{1}{2}\left(\alpha\left(\frac{1}{2}\right) + \beta\left(\frac{1}{2}\right)\right) &\iff u - \beta\left(\frac{1}{2}\right) \leq \frac{1}{2}\left(\alpha\left(\frac{1}{2}\right) - \beta\left(\frac{1}{2}\right)\right) \\ &\iff \frac{u - \beta\left(\frac{1}{2}\right)}{\alpha\left(\frac{1}{2}\right) - \beta\left(\frac{1}{2}\right)} \leq \frac{1}{2}. \end{aligned}$$

We now convert the relation in functions into relation in sets. After divide $B_{1/2}$ into two portions, we deduce either

$$m\left(\left\{u \geq \frac{1}{2}\left(\alpha\left(\frac{1}{2}\right)+\beta\left(\frac{1}{2}\right)\right)\bigcap B_{1/2}\right\}\right)=m\left(\left\{\frac{\alpha\left(\frac{1}{2}\right)-u}{\alpha\left(\frac{1}{2}\right)-\beta\left(\frac{1}{2}\right)} \leq \frac{1}{2}\right\}\bigcap B_{1/2}\right) \geq \frac{1}{2}m\left(B_{1/2}\right) \tag{1.2.10}$$

or

$$m\left(\left\{u \leq \frac{1}{2}\left(\alpha\left(\frac{1}{2}\right)+\beta\left(\frac{1}{2}\right)\right)\bigcap B_{1/2}\right\}\right)=m\left(\left\{\frac{u-\beta\left(\frac{1}{2}\right)}{\alpha\left(\frac{1}{2}\right)-\beta\left(\frac{1}{2}\right)} \leq \frac{1}{2}\right\}\bigcap B_{1/2}\right) \geq \frac{1}{2}m\left(B_{1/2}\right) \tag{1.2.11}$$

holds, otherwise there exists $x_0 \in B_{1/2}$, such that

$$1<\frac{\alpha\left(\frac{1}{2}\right)-u(x_0)}{\alpha\left(\frac{1}{2}\right)-\beta\left(\frac{1}{2}\right)}+\frac{u(x_0)-\beta\left(\frac{1}{2}\right)}{\alpha\left(\frac{1}{2}\right)-\beta\left(\frac{1}{2}\right)}=1.$$

Base on (1.2.10) or (1.2.11), we may apply Weak Harnack Theorem 1.2.1. To elaborate, if (1.2.10) holds, define a function

$$w(x):=\frac{\alpha\left(\frac{1}{2}\right)-u(x)}{\alpha\left(\frac{1}{2}\right)-\beta\left(\frac{1}{2}\right)}-\frac{1}{2},$$

since u is a solution (certainly be a supersolution), hence w is a subsolution. Simultaneously, $-1/2 \leq w \leq 1/2$ holds $\forall\, x \in B_{1/2}$ implies that $\|w^+\|_{L^\infty(B_{1/2})} \leq 1/2$. Then theorem 1.2.1 implies that there exists $C(1/2)>0$ s.t.

$$\left\|\left(\frac{\alpha\left(\frac{1}{2}\right)-u}{\alpha\left(\frac{1}{2}\right)-\beta\left(\frac{1}{2}\right)}-\frac{1}{2}\right)^+\right\|_{L^\infty(B_{1/4})} \leq \frac{1}{2}-\frac{1}{2}C\left(\frac{1}{2}\right)$$

Therefore, for $x \in B_{1/4}$, either

$$\alpha\left(\frac{1}{2}\right)-u(x) \leq \frac{1}{2}\left(\alpha\left(\frac{1}{2}\right)-\beta\left(\frac{1}{2}\right)\right)$$

or

$$\alpha\left(\frac{1}{2}\right)-u(x) \leq \left(1-\frac{1}{2}C\left(\frac{1}{2}\right)\right)\left(\alpha\left(\frac{1}{2}\right)-\beta\left(\frac{1}{2}\right)\right).$$

Clearly, this inequality holds for a.e. $x \in \{\omega(x)>0\}$. Denote a constant γ as the greater of $1/2$ and $(1-1/2C\left(1/2\right))$, above estimates read

$$\mathrm{osc}_{B_{1/4}}u=\alpha\left(\frac{1}{4}\right)-\beta\left(\frac{1}{4}\right) \leq \alpha\left(\frac{1}{2}\right)-\beta\left(\frac{1}{4}\right) \leq \gamma\left(\alpha\left(\frac{1}{2}\right)-\beta\left(\frac{1}{2}\right)\right)=\gamma\,\mathrm{osc}_{B_{1/2}}u.$$

Alternatively, suppose (1.2.11) holds, apply Theorem 1.2.1 to the function

$$w := \frac{u(x) - \beta\left(\frac{1}{2}\right)}{\alpha\left(\frac{1}{2}\right) - \beta\left(\frac{1}{2}\right)} - \frac{1}{2}$$

for the same $C(1/2) > 0$, we have

$$\left\| \left(\frac{u - \beta\left(\frac{1}{2}\right)}{\alpha\left(\frac{1}{2}\right) - \beta\left(\frac{1}{2}\right)} - \frac{1}{2} \right)^{+} \right\|_{L^\infty(B_{1/4})} \leq \frac{1}{2} - \frac{1}{2} C\left(\frac{1}{2}\right).$$

Correspondingly, for $x \in B_{1/4}$, either

$$u(x) - \beta\left(\frac{1}{2}\right) \leq \frac{1}{2}\left(\alpha\left(\frac{1}{2}\right) - \beta\left(\frac{1}{2}\right)\right)$$

or

$$u(x) - \beta\left(\frac{1}{2}\right) \leq \left(1 - \frac{1}{2} C\left(\frac{1}{2}\right)\right)\left(\alpha\left(\frac{1}{2}\right) - \beta\left(\frac{1}{2}\right)\right).$$

Then

$$\operatorname{osc}_{B_{1/4}} u = \alpha\left(\frac{1}{4}\right) - \beta\left(\frac{1}{4}\right) \leq \alpha\left(\frac{1}{4}\right) - \beta\left(\frac{1}{2}\right) \leq \gamma\left(\alpha\left(\frac{1}{2}\right) - \beta\left(\frac{1}{2}\right)\right) = \gamma \operatorname{osc}_{B_{1/2}} u.$$

To summarize, in both cases the constant $\gamma \in (0,1)$ we constructed established the oscillation decay (1.2.9).

□

As a corollary, we can prove the following

Corollary 1.2.4 (From L^∞ to C^α)**.** *Assume nonnegative, bounded function $u \in H^1(B_1)$ solves $\mathcal{L}u = 0$ in B_1 in the weak sense. Then $u \in C^\alpha(B_{1/4})$ for some $\alpha = \alpha(n, \lambda, \Lambda) \in (0,1)$. In particular*

$$\|u\|_{C^\alpha(B_{1/4})} \leq C\|u\|_{L^\infty(B_{1/2})}. \tag{1.2.12}$$

Proof. Local Boundedness Theorem gives $u \in L^\infty(B_{1/2})$. Since the regularity properties of the solution remain invariant under linear scaling, W.L.O.G. assume $\|u\|_{L^\infty(B_{1/2})} \leq \frac{1}{2}$. Then $\operatorname{osc}_{B_{1/2}} u = \sup_{B_{1/2}} u - \inf_{B_{1/2}} u \leq 1$. To prove (1.2.12), we will show that there exists $\alpha \in (0,1)$ s.t. for some $C \geq 0$,

$$\sup_{x,y \in B_{1/4}, x \neq y} \frac{|u(x) - u(y)|}{|x - y|^\alpha} \leq C \tag{1.2.13}$$

holds.

Take any $y \in B_{1/4}$, notice that for any $k \geq 2$, $B_{2^{-k}}(y) \subset B_{1/2}$. Consider arbitrary $x \in B_{2^{-k}}(y) \setminus B_{2^{-(k+1)}}(y)$, use the oscillation decay introduced in theorem 1.2.3, we have

$$\begin{aligned}|u(x) - u(y)| &\leq \operatorname{osc}_{B_{2^{-k}}(y)} u \leq \gamma \operatorname{osc}_{B_{2^{-(k-1)}}(y)} u \\ &\leq \cdots \leq \gamma^{k-2} \operatorname{osc}_{B_{1/4}(y)} u \leq \gamma^{k-2} \operatorname{osc}_{B_{1/2}} u \leq \gamma^{k-2}\end{aligned}$$

We rename

$$\gamma^{k-2} = 2^{(k-2)\log_2 \gamma} = C_0 2^{-\alpha k} \quad \alpha := -\log_2 \gamma \in (0,1),\ C_0 := \gamma^{-2}$$

and notice that for our choice of x and y

$$|x - y| \geq 2^{-(k+1)} \implies |x-y|^\alpha \geq 2^{-(k+1)\alpha} = \gamma^{k+1},$$

thus $\forall\, y \in B_{1/4}, x \in B_{2^{-k}}(y) \setminus B_{2^{-(k+1)}}(y)$,

$$|u(x) - u(y)| \leq \gamma^{k-2} \leq \gamma^{-3}|x-y|^\alpha$$

holds. Divide to the LHS, taking supremum first in k, then in y to conclude (1.2.13).

□

3 Gradient Estimates for Some Equations

Part I: gradient estimates for harmonic equation and heat equation

Theorem 1.3.1. *Let M be a complete Riemannian manifold with $\dim M = n \geq 2$ and $\operatorname{Ric}(M) \geq -(n-1)k$ ($k \geq 0$). Suppose u is a positive harmonic function on M, then for any geodesic ball $B_a(x)$ in M, we have*

$$\frac{|\nabla u|}{u} \leq C_n \left(\frac{1 + a k^{\frac{1}{2}}}{a} \right), \tag{1.3.1}$$

where C_n is a constant depending only on n.

Theorem 1.3.2. *Let M be a compact Riemannian manifold possibly with boundary, $\mathrm{Ric}(M) \geq 0$, and $u(x,t)$ be a non-negative solution to the heat equation on M:*

$$\left(\Delta - \frac{\partial}{\partial t}\right) u(x,t) = 0. \tag{1.3.2}$$

When $\partial M \neq \emptyset$, assume that ∂M is convex, i.e., the second fundamental form $\Pi \geq 0$. In addition, $u(x,t)$ satisfies the Neumann boundary condition:

$$\frac{\partial u}{\partial \nu} = 0 \quad \text{on } \partial M \times (0,\infty),$$

where $\frac{\partial}{\partial \nu}$ denotes the outward normal direction of ∂M. Then on $M \times (0,\infty)$, u satisfies the following gradient estimate:

$$\frac{|\nabla u|^2}{u^2} - \frac{u_t}{u} \leq \frac{n}{2t}. \tag{1.3.3}$$

Theorem 1.3.3. *Let M be a complete Riemannian manifold with $\mathrm{Ric}(M) \geq -k$, and let B_{2R} be the geodesic ball centered at a fixed point O with radius $2R$. Suppose $u(x,t)$ is a positive solution to the heat equation on $B_{2R} \times [0,\infty]$:*

$$\left(\Delta - \frac{\partial}{\partial t}\right) u(x,t) = 0. \tag{1.1}$$

For $\alpha > 1$, the following gradient estimate holds in B_R:

$$\begin{aligned}\sup_{B_R}\left(\frac{|\nabla u|^2}{u^2} - \alpha\frac{u_t}{u}\right) \leq \frac{C\alpha^2}{R^2}\left(\frac{\alpha^2}{\alpha-1} + \sqrt{k}R\right)\\ + \frac{n\alpha^2 k}{2(\alpha-1)} + \frac{n\alpha^2}{2t},\end{aligned}$$

where C is a constant depending only on n.

Theorem 1.3.4 (Yau, Li-Wang)**.** *Let M^m be a complete manifold with Ricci curvature bounded from below by*

$$Ric_M \geq -(m-1)K$$

for some constant $K \geq 0$. If f is a positive function defined on the geodesic ball $B_p(2R) \subset M$ satisfying $\Delta f = -\lambda f$ for some constant $\lambda \geq 0$, then there exists a constant C depending on m such that

$$\frac{|\nabla f|^2}{f^2}(x) \leq \frac{(4(m-1)^2 + 2\epsilon)K}{4 - 2\epsilon} + C((1+\epsilon^{-1})R^{-2} + \lambda),$$

for all $x \in B_p(R)$ *and for any* $\epsilon < 2$.

Moreover, if f *is defined on* M*, then*

$$\frac{|\nabla f|^2}{f^2}(x) \leq \frac{(m-1)^2 K}{2} - \lambda + \sqrt{\frac{(m-1)^4 K^2}{4} - (m-1)^2 \lambda K}.$$

and

$$\lambda \leq \frac{(m-1)^2 K}{4}.$$

Proof. $\square$ $\square$

Corollary 1.3.5 (A Harnack Type Inequality)**.** *Let* M^m *be a complete manifold with Ricci curvature bounded from below by*

$$Ric_M \geq -(m-1)K$$

for some constant $K \geq 0$*. If* f *is a positive function defined on the geodesic ball* $B_p(2R) \subset M$ *satisfying* $\Delta f = -\lambda f$ *for some constant* $\lambda \geq 0$*, then there exists constants* C_9*,* $C_{10} > 0$ *depending on* m *such that*

$$f(x) \leq f(y) C_9 \exp(C_{10} R \sqrt{K + \lambda})$$

for all $x, y \in B_p\left(\frac{R}{2}\right)$.

Corollary 1.3.6 (Cheng)**.** *Let* M^m *be a complete manifold with Ricci curvature satisfying*

$$Ric_M \geq -(m-1)K,$$

for some constant $K \geq 0$*. If we denote* $\lambda_1(M)$ *to be the infimum of the spectrum of the Laplacian acting on* L^2 *functions, then*

$$\lambda_1(M) \leq \frac{(m-1)^2}{4}.$$

4 Decay Estimates for Linear Elliptic Equations on Cylinder

In this section, we focus on a decay estimate for solutions to a linear elliptic equation on cylinder, which carried out by prof. Yin Hao in various methods. We will illustrate each version in detail.

First version: via the separation of variables approach

Consider a vector bundle E over S^3 equipped with an inner product, and let L be a self-adjoint, non-negative, elliptic operator defined on sections of E. Let $u(t)$ and $f(t)$ be two families of sections of E, satisfying

$$(\partial_t^2 - L)u(t) = f(t) \tag{1.3.1}$$

Recall some properties of self-adjoint elliptic operators on compact manifolds:

- Their spectrum is discrete, i.e., there are countably many eigenvalues;
- The eigenvalues tend to infinity;
- The multiplicity of each eigenvalue is finite.

List the eigenvalues of L in increasing order as $\lambda_1^2, \lambda_2^2, \cdots, \lambda_n^2, \cdots$, and denote the corresponding L^2-normalized eigenfunctions by φ_i. These eigenvalues λ_i are listed with multiplicities and are non-negative. We now state and prove the main theorem on decay estimate:

Theorem 1.4.1. *Suppose α is a positive number different from any λ_i. If f is a smooth function satisfying $|f| \le e^{-\alpha M} e^{\alpha|t|}$ for $t \in [-M, M]$, then there exists a smooth solution $u(t)$ to* (1.3.1) *such that*

$$|u| \le Ce^{-\alpha M} e^{\alpha|t|}, \quad \textit{for} \quad t \in [-M, M]. \tag{1.3.2}$$

Here, C depends on α, but not on M.

Proof. First, expand $f(t)$ in terms of the eigenfunctions:

$$f(t) = \sum_{i=1}^{\infty} a_i(t)\varphi_i.$$

By the Parseval inequality, we have

$$\sum_{i=1}^{\infty} a_i^2(t) \le Ce^{-2\alpha M} e^{2\alpha|t|}. \tag{1.3.3}$$

Formally, if the solution u to equation (1.27) can be expanded as:

$$u(t) = \sum_{i=1}^{\infty} A_i(t)\varphi_i,$$

then each A_i satisfies the following ODE:

$$\partial_t^2 A_i - \lambda_i^2 A_i = a_i. \tag{1.3.4}$$

The idea of the proof is: for each i, we construct a suitable A_i, then verify the convergence of the series $\sum_i A_i$, and finally obtain the solution to equation (1.3.1).

Step 1. For each λ_i, we find a solution A_i to equation (1.3.4), though the solution may not be unique. The construction of A_i depends on the value of λ_i.

Case 1. $\lambda_1 = 0$. In this case, equation (1.3.4) becomes $\partial_t^2 A_1 = a_1$, we choose the solution

$$A_1(t) = \int_0^t \int_0^s a_1(x)\, dx\, ds.$$

By (1.29), we have

$$\begin{aligned}
A_1^2(t) &\le \left(\int_0^{|t|} \int_0^{|s|} |a_1(x)|\, dx\, ds \right)^2 \\
&\le C \left(\int_0^{|t|} \int_0^{|s|} e^{-\alpha M} e^{\alpha|x|}\, dx\, ds \right)^2 \\
&= C e^{-2\alpha M} \left(\int_0^{|t|} \frac{e^{\alpha|s|} - 1}{\alpha}\, ds \right)^2 \\
&= C e^{-2\alpha M} \left(e^{\alpha|t|} - 1 - \alpha|t| \right)^2 \\
&\le C e^{-2\alpha M} e^{2\alpha|t|}.
\end{aligned}$$

Case 2. $\lambda_i < \alpha$. There are only finitely many such λ_i. For $t \in [-M, M]$, a solution to equation (1.3.4) is

$$A_i(t) = e^{\lambda_i t} \int_0^t e^{-2\lambda_i s} \left(\int_0^s a_i(x) e^{\lambda_i x}\, dx \right) ds,$$

and for $t > 0$, we estimate

$$\begin{aligned}
|A_i(t)| &\le e^{\lambda_i t} \int_0^t e^{-2\lambda_i s} \left(\int_0^s |a_i(x)| e^{\lambda_i x} dx \right) ds \\
&\le C e^{-\alpha M} e^{\lambda_i t} \int_0^t e^{-2\lambda_i s} \left(\int_0^s e^{(\alpha+\lambda_i)x} dx \right) ds \\
&= \frac{C}{\alpha+\lambda_i} e^{-\alpha M} e^{\lambda_i t} \int_0^t e^{-2\lambda_i s} \left(e^{(\alpha+\lambda_i)s} - 1 \right) ds \\
&= \frac{C}{\alpha+\lambda_i} e^{-\alpha M} \left(\frac{e^{\alpha t} - e^{\lambda_i t}}{\alpha - \lambda_i} + \frac{e^{-\lambda_i t} - e^{\lambda_i t}}{2\lambda_i} \right) \\
&\le C e^{-\alpha M} e^{\alpha t}.
\end{aligned}$$

Similarly, for $t < 0$, let $\tilde{t} = -t$ and we obtain

$$\begin{aligned}
|A_i(t)| &= \left| e^{-\lambda_i \tilde{t}} \int_0^{\tilde{t}} e^{2\lambda_i s} \left(\int_0^s a_i(-x) e^{-\lambda_i x} dx \right) ds \right| \\
&\le e^{-\lambda_i \tilde{t}} \int_0^{\tilde{t}} e^{2\lambda_i s} \left(\int_0^s |a_i(-x)| e^{-\lambda_i x} dx \right) ds \\
&\le C e^{-\alpha M} e^{-\lambda_i \tilde{t}} \int_0^{\tilde{t}} e^{2\lambda_i s} \left(\int_0^s e^{(\alpha-\lambda_i)x} dx \right) ds \\
&= \frac{C}{\alpha-\lambda_i} e^{-\alpha M} e^{-\lambda_i \tilde{t}} \int_0^{\tilde{t}} \left(e^{(\alpha+\lambda_i)s} - e^{2\lambda_i s} \right) ds \\
&= \frac{C}{\alpha-\lambda_i} e^{-\alpha M} \left(\frac{e^{\alpha \tilde{t}} - e^{-\lambda_i \tilde{t}}}{\alpha + \lambda_i} - \frac{e^{\lambda_i \tilde{t}} - e^{-\lambda_i \tilde{t}}}{2\lambda_i} \right) \\
&\le C e^{-\alpha M} e^{\alpha |t|}.
\end{aligned}$$

Case 3. $\lambda_i > \alpha$. Observe that the following is also a solution to equation (1.3.4):

$$A_i(t) = -\frac{1}{2\lambda_i} e^{\lambda_i t} \int_t^\infty e^{-\lambda_i x} a_i(x)\, dx - \frac{1}{2\lambda_i} e^{-\lambda_i t} \int_{-\infty}^t e^{\lambda_i x} a_i(x)\, dx.$$

We extend the domain of a_i by setting $a_i = 0$ for $|x| > M$. Then,

$$\begin{aligned}
|A_i|^2(t) &\le \frac{1}{2\alpha^2} e^{2\lambda_i t} \left(\int_t^\infty e^{-\lambda_i x} a_i(x) dx \right)^2 + \frac{1}{2\alpha^2} e^{-2\lambda_i t} \left(\int_{-\infty}^t e^{\lambda_i x} a_i(x) dx \right)^2 \\
&:= I + II.
\end{aligned}$$

We estimate I and II separately, W.L.O.G. assuming $t > 0$, and choose λ' and δ such that for all $\lambda_i > \alpha$, we have $\alpha + \delta < \lambda' < \lambda_i$. Then,

$$\begin{aligned}
I &\leq C(\alpha)e^{2\lambda_i t}\left(\int_t^\infty e^{-2\delta x}dx\right)\left(\int_t^\infty e^{2\delta x}e^{-2\lambda_i x}a_i^2(x)dx\right)\\
&\leq C(\alpha,\delta)e^{-2\delta t}\int_t^\infty e^{2\delta x}e^{2\lambda_i(t-x)}a_i^2(x)dx\\
&\leq C(\alpha,\delta)e^{-2\delta t}\int_t^\infty e^{2\delta x}e^{2\lambda'(t-x)}a_i^2(x)dx.
\end{aligned}$$

For II,

$$\begin{aligned}
II &\leq C(\alpha)e^{-2\lambda_i t}\left(\int_{-\infty}^0 e^{\lambda_i x}a_i(x)dx\right)^2 + C(\alpha)e^{-2\lambda_i t}\left(\int_0^t e^{\lambda_i x}a_i(x)dx\right)^2\\
&\leq C(\alpha,\delta)e^{-2\lambda_i t}\int_{-\infty}^0 e^{-2\delta x}e^{2\lambda_i x}a_i^2(x)dx\\
&\quad + C(\alpha,\delta)e^{-2\lambda_i t}e^{2\delta t}\int_0^t e^{-2\delta x}e^{2\lambda_i x}a_i^2(x)dx\\
&\leq C(\alpha,\delta)\int_{-\infty}^0 e^{-2\delta x}e^{-2\lambda'(t-x)}a_i^2(x)dx\\
&\quad + C(\alpha,\delta)e^{2\delta t}\int_0^t e^{-2\delta x}e^{-2\lambda'(t-x)}a_i^2(x)dx.
\end{aligned}$$

Step 2. Verify that the A_i determined in Step 1 ensure the convergence of the series $\sum_i A_i(t)\varphi_i$.

Since only finitely many λ_i are less than α,

$$\begin{aligned}
\sum_i A_i^2(t) &= \sum_{\lambda_i<\alpha} A_i^2(t) + \sum_{\lambda_i>\alpha} A_i(t)\\
&\leq Ce^{-2\alpha M}e^{-2\alpha|t|} + \sum_i (I + II).
\end{aligned}$$

We estimate $\sum_i I$ and $\sum_i II$ respectively:

$$\begin{aligned}
\sum_i I &\leq C(\alpha,\delta)e^{-2\delta t}\int_t^\infty e^{2\delta x}e^{2\lambda'(t-x)}\sum_i a_i^2(x)dx\\
&\leq Ce^{-2\alpha M}e^{-2\delta t}\int_t^\infty e^{2\delta x}e^{2\lambda'(t-x)}e^{2\alpha x}dx\\
&= \frac{C}{2(\lambda'-\delta-\alpha)}e^{-2\alpha M}e^{-2\delta t}e^{2\lambda' t}e^{-2(\lambda'-\delta-\alpha)t}\\
&= Ce^{-2\alpha M}e^{2\alpha t},
\end{aligned}$$

Similarly,

$$
\begin{aligned}
\sum_i II &\le C(\alpha,\delta)\int_{-\infty}^{0} e^{-2\delta x}e^{-2\lambda'(t-x)}\sum_i a_i^2(x)dx \\
&\quad + C(\alpha,\delta)e^{2\delta t}\int_0^t e^{-2\delta x}e^{-2\lambda'(t-x)}\sum_i a_i^2(x)dx \\
&\le Ce^{-2\alpha M}\int_{-\infty}^{0} e^{-2\delta x}e^{-2\lambda'(t-x)}e^{2\alpha|x|}dx \\
&\quad + Ce^{-2\alpha M}e^{2\delta t}\int_0^t e^{-2\delta x}e^{-2\lambda'(t-x)}e^{2\alpha x}dx \\
&= \frac{C}{2(\lambda'-\delta-\alpha)}e^{-2\alpha M}e^{-2\lambda' t} \\
&\quad + \frac{C}{2(\alpha+\lambda'-\delta)}e^{-2\alpha M}e^{2\delta t}e^{-2\lambda' t}(e^{2(\alpha+\lambda'-\delta)t}-1) \\
&\le \frac{C}{2(\lambda'-\delta-\alpha)}e^{-2\alpha M}\left(e^{-2\lambda' t}+e^{2\alpha t}-e^{2(\delta-\lambda')t}\right) \\
&\le Ce^{-2\alpha M}e^{2\alpha t}.
\end{aligned}
$$

Finally, the above estimate implies that $u_N := \sum_{i=1}^N A_i(t)\varphi_i$ converges to $u = \sum_{i=1}^\infty A_i(t)\varphi_i$ locally in L^2. Moreover, u_N is a classical solution to the equation $(\partial_t^2 - L)u_N = f_N$, where $f_N := \sum_{i=1}^N a_i(t)\varphi_i$. Since f_N converges to f smoothly as $N\to\infty$, the elliptic estimates imply that u_N converges to u smoothly. Hence u is a classical solution to (1.3.1).

$\square$

Second version:

5 Approach Based on Conservation Laws

Theorem 1.5.1 (Rivière)**.** *Let* $\Omega\in L^2(D^2, so(m)\otimes\Lambda^1 D^2)$*, assume* $A, B\in W^{1,2}(D^2, M(m))$ *and satisfy*

$$\nabla A - A\Omega = -\nabla^\perp B. \tag{4.11}$$

If the matrix A is almost everywhere invertible, and

$$\|A\|_{L^\infty(D^2)} + \|A^{-1}\|_{L^\infty(D^2)} < +\infty, \tag{4.12}$$

then u is a solution to the Schrödinger system (4.6) if and only if u satisfies the conservation law

$$\operatorname{div}(A\nabla u - B\nabla^{\perp} u) = 0. \tag{4.13}$$

6 Differential Inequalities and Applications

Theorem 1.6.1. $V(x)$ *is a continuously differentiable positive definite function. If there exists* $\dot{V}(x) + aV(x) + bV^{\alpha}(x) \leq 0$*, where* a, b *are positive constants and* $0 < \alpha < 1$*, then* $V(x)$ *converges to zero in finite time, and the convergence time* T *satisfies*

$$T \leq \frac{1}{a(1-\alpha)} \ln \frac{aV^{1-\alpha}(x_0) + b}{b}.$$

Chapter 2 The Variational Approach and Its Applications

1 Pohozaev Identity

Let X be a Banach space and let $T(t) : X \to X$ be a family of transformations such that $T(1) = \mathrm{id}$. If $u \in X$ is a critical point of the functional $\varphi \in \mathcal{C}^1(X, \mathbb{R})$ (this symbol means the Fréchet derivate of ϕ exists and is continuous on X), we conjecture that

$$\frac{\partial}{\partial t}\Big|_{t=1} \varphi(T(t)u) = 0. \tag{2.1.1}$$

Before we consider an example, recall

$$\mathcal{D}^{1,2}(\mathbb{R}^N) := \{u \in L^{2^*}(\mathbb{R}^N) : \nabla u \in L^2(\mathbb{R}^N)\}$$

where $N \geq 3$ and $2^* := 2N/(N-2)$, with the inner product $\int_{\mathbb{R}^N} \nabla u \cdot \nabla v$ and the corresponding norm $\left(\int_{\mathbb{R}^N} |\nabla u|^2\right)^{1/2}$ is a Hilbert space. Moreover, the space $\mathcal{D}_0^{1,2}(\Omega)$ is the closure of $C_c^\infty(\Omega)$ in $\mathcal{D}^{1,2}(\mathbb{R}^N)$. For simplicity of notations, we shall write $2^* = \infty$ when $N = 1$ or $N = 2$.

Now denote $X := D^{1,2}(\mathbb{R}^N)$, $T(t)u(x) := u(x/t), t > 0$ and $\varphi(u) := \int_{\mathbb{R}^N} (|\nabla u|^2/2 - F(u))$, we have

$$\varphi(T(t)u) = \frac{t^{N-2}}{2} \int_{\mathbb{R}^N} |\nabla u|^2 - t^N \int_{\mathbb{R}^N} F(u)$$

and

$$\frac{\partial}{\partial t}\Big|_{t=1} \varphi(T(t)u) = \frac{N-2}{2} \int_{\mathbb{R}^N} |\nabla u|^2 - N \int_{\mathbb{R}^N} F(u).$$

If u is a critical point of φ, we conjecture that

$$\frac{N-2}{2} \int_{\mathbb{R}^N} |\nabla u|^2 = N \int_{\mathbb{R}^N} F(u).$$

Part 1: Bounded domains

We first consider an elliptic equation on bounded domain:

$$(\mathcal{P}_1)\quad \begin{cases} -\Delta u = f(u), \\ \quad u \in H_0^1(\Omega), \end{cases} \tag{2.1.2}$$

where $f \in C^1(\mathbb{R}, \mathbb{R})$ and Ω is a smooth bounded domain of $\mathbb{R}^N$, $N \geq 3$.

The action of $t > 0$ is defined by $T(t)u(x) := u(x/t)$ and the corresponding generator is $\frac{\partial}{\partial t}\big|_{t=1} T(t) = -x \cdot \nabla$. In order to prove the Pohozaev identity, we multiply $-\Delta u = f(u)$ by $x \cdot \nabla u$ and integrate by parts. We define $F(u) := \int_0^u f(s)ds$.

Theorem 2.1.1 (Pohozaev identity, 1965)**.** *Let $u \in H^2_{loc}(\bar{\Omega})$ be a solution of $(\mathcal{P}_1)$ such that $F(u) \in L^1(\Omega)$. Then u satisfies*

$$\frac{1}{2}\int_{\partial\Omega} |\nabla u|^2 x \cdot \nu \, d\sigma = N \int_\Omega F(u)\, dx - \frac{N-2}{2}\int_\Omega |\nabla u|^2 dx, \tag{2.1.3}$$

where ν denotes the unit outward normal to $\partial\Omega$.

Proof. It follows from $(\mathcal{P}_1)$ that $0 = (\Delta u + f(u))x \cdot \nabla u$. Compute each term respectively,

$$\begin{aligned} f(u)x \cdot \nabla u &= \sum_{i=1}^N x_i f(u)\partial_i u \\ &= \sum_{i=1}^N x_i \partial_i F(u) \\ &= \operatorname{div}(xF(u)) - NF(u), \end{aligned}$$

For $\Delta u x \cdot \nabla u$, notice that

$$\begin{aligned} \operatorname{div}\big((x \cdot \nabla u)\nabla u\big) &= (x \cdot \nabla u)\operatorname{div}\nabla u + \nabla(x \cdot \nabla u) \cdot \nabla u \\ &= (x \cdot \nabla u)\Delta u + |\nabla u|^2 + \sum_{k,l} u_{kl} x_k u_l \\ &= (x \cdot \nabla u)\Delta u + |\nabla u|^2 + \frac{1}{2} x \cdot \nabla(|\nabla u|^2), \end{aligned}$$

thus

$$\begin{aligned} \Delta u(x \cdot \nabla u) &= \operatorname{div}\big((x \cdot \nabla u)\nabla u\big) - |\nabla u|^2 - \frac{1}{2} x \cdot \nabla(|\nabla u|^2) \\ &= \operatorname{div}\left((x \cdot \nabla u)\nabla u - \frac{1}{2} x |\nabla u|^2\right) + \frac{N-2}{2}|\nabla u|^2. \end{aligned}$$

Combine above results with equation $0 = (\Delta u + f(u))x \cdot \nabla u$ yields

$$NF(u) - \frac{N-2}{2}|\nabla u|^2 = \operatorname{div}\left(xF(u) + (x \cdot \nabla u)\nabla u - \frac{1}{2}x|\nabla u|^2\right),$$

By Divergence Theorem,

$$\begin{aligned}\int_\Omega \left(NF(u) - \frac{N-2}{2}|\nabla u|^2\right) dx &= \int_\Omega \operatorname{div}\left(xF(u) + (x \cdot \nabla u)\nabla u - \frac{1}{2}x|\nabla u|^2\right) dx \\ &= \int_{\partial\Omega} \left(xF(u) + (x \cdot \nabla u)\nabla u - \frac{1}{2}x|\nabla u|^2\right) \cdot \nu d\sigma.\end{aligned}$$

However, on $\partial\Omega$, we have $u = 0$ (which implies $F(u) = 0$) and $\nabla u = \frac{\partial u}{\partial \nu}\nu$, therefore,

$$\begin{aligned}\int_\Omega \left(NF(u) - \frac{N-2}{2}|\nabla u|^2\right) dx &= \int_{\partial\Omega} \left((x \cdot v)\left|\frac{\partial u}{\partial \nu}\right|^2 - \frac{1}{2}\left|\frac{\partial u}{\partial \nu}\right|^2 (x \cdot \nu)\right) d\sigma \\ &= \frac{1}{2}\int_{\partial\Omega} |\nabla u|^2 x \cdot \nu d\sigma.\end{aligned}$$

□

Immediately,

Corollary 2.1.2 (Rellich identity, 1940). *Let $u \in H^2_{loc}(\tilde{\Omega})$ be a solution of*

$$\begin{cases} -\Delta u = \lambda u, \\ \quad u \in H^1_0(\Omega). \end{cases}$$

Then u satisfies

$$\frac{1}{2}\int_{\partial\Omega} |\nabla u|^2 x \cdot \nu d\sigma = \int_\Omega \lambda u^2 dx.$$

Proof. It suffices to use the Pohozaev identity since

$$\int_\Omega F(u)dx = \frac{\lambda}{2}\int_\Omega u^2 dx, \int_\Omega |\nabla u|^2 dx = \lambda \int_\Omega u^2 dx.$$

□

Part 2: Unbounded domains

We now consider the problem

$$(\mathcal{P}_2) \begin{cases} -\Delta u = f(u), \\ \quad u \in \mathcal{D}^{1,2}_0(\Omega), \end{cases} \tag{2.1.4}$$

where $f \in C^1(\mathbb{R}, \mathbb{R}), f(0) = 0$, and Ω is a smooth unbounded domain of $\mathbb{R}^N$, $N \geq 3$.

In order to prove the Pohozaev identity, we use a truncation argument due to Kavian to get a similar result as theorem 2.1.1. To reduce the proof on an unbounded domain to that on a bounded domain, we choose a cut-off function. The proof process does not involve any essential differences.

Theorem 2.1.3. *Let $u \in H^q_{loc}(\tilde{\Omega})$ be a solution of $(\mathcal{P}_2)$ such that $F(u) \in L^1(\Omega)$. Then the Pohozaev identity is valid.*

Proof. Let $\psi \in C_c^\infty(\Omega)$ be such that $0 \leq \psi \leq 1$, $\psi(r) = 1$ for $r \leq 1$ and $\psi(r) = 0$ for $r \geq 2$. Define $\psi_n(x) := \psi(|x|^2/n^2)$ on $\mathbb{R}^N$, then there exists $C \geq 0$ such that for every n, $\psi_n \leq C, |x||\nabla\psi_n(x)| \leq C$.

It follows from $(\mathcal{P}_2)$ that, for every n, $0 = (\Delta u + f(u))\psi_n x \cdot \nabla u$ holds, we still compute each term respectively.

$$\begin{aligned} f(u)\psi_n(x \cdot \nabla u) &= \sum_k x_k f(u)\psi_n u_k \\ &= \operatorname{div}(\psi_n F(u)x) - N\psi_n F(u) - F(u)(x \cdot \nabla\psi_n). \end{aligned}$$

For $\Delta u \psi_n(x \cdot \nabla u)$, notice that

$$\begin{aligned} \operatorname{div}(\psi_n(x \cdot \nabla u)\nabla u) &= \psi_n(x \cdot \nabla u)\operatorname{div}(\nabla u) + \nabla\big(\psi_n(x \cdot \nabla u)\big)\nabla u \\ &= \psi_n(x \cdot \nabla u)\Delta u + u_l\big(\psi_n(x_k u_k)\big)_l \\ &= \psi_n(x \cdot \nabla u)\Delta u + u_l\big((\psi_n)_l(x_k u_k) + \psi_n(x_k u_{kl} + \delta_{kl}u_k)\big) \\ &= \psi_n(x \cdot \nabla u)\Delta u + (\nabla u \cdot \nabla\psi_n)(x \cdot \nabla u) + \frac{1}{2}\psi_n(x \cdot \nabla|\nabla u|^2) + \psi_n|\nabla u|^2, \end{aligned}$$

thus

$$\begin{aligned} \psi_n(x \cdot \nabla u)\Delta u &= \operatorname{div}(\psi_n(x \cdot \nabla u)\nabla u) - (\nabla u \cdot \nabla\psi_n)(x \cdot \nabla u) - \frac{1}{2}\psi_n(x \cdot \nabla|\nabla u|^2) - \psi_n|\nabla u|^2 \\ &= \operatorname{div}\left(\psi_n\left((x \cdot \nabla u)\nabla u - \frac{1}{2}|\nabla u|^2 x\right)\right) + \frac{N-2}{2}\psi_n|\nabla u|^2 \\ &\quad - (\nabla u \cdot \nabla\psi_n)(x \cdot \nabla u) + \frac{1}{2}|\nabla u|^2(\nabla\psi_n \cdot x) \end{aligned}$$

Combine above results with equation $0 = (\Delta u + f(u))\psi_n x \cdot \nabla u$ yields

$$\begin{aligned}&\operatorname{div}\left(\psi_n\left(F(u)x + (x\cdot\nabla u)\nabla u - \frac{1}{2}|\nabla u|^2 x\right)\right)\\&= N\psi_n F(u) + F(u)(x\cdot\nabla\psi_n) - \frac{N-2}{2}\psi_n|\nabla u|^2\\&\quad + (\nabla u\cdot\nabla\psi_n)(x\cdot\nabla u) - \frac{1}{2}|\nabla u|^2(\nabla\psi_n\cdot x)\end{aligned}$$

By Divergence Theorem, for every n,

$$\begin{aligned}&\int_{\partial\Omega}\psi_n\left(F(u)x + (x\cdot\nabla u)\nabla u - \frac{1}{2}|\nabla u|^2 x\right)\cdot\nu d\sigma\\&= \int_\Omega\left(\left(NF(u) - \frac{N-2}{2}|\nabla u|^2\right)\psi_n + F(u)(x\cdot\nabla\psi_n)\right.\\&\quad\left.-\frac{1}{2}|\nabla u|^2(\nabla\psi_n\cdot x) + (\nabla u\cdot\nabla\psi_n)(x\cdot\nabla u)\right)dx.\end{aligned}$$

The Lebesgue Dominated Convergence Theorem implies that (we have used the facts: for every n, $\psi_n \leq C, |x||\nabla\psi_n(x)| \leq C$ and $u \in \mathcal{D}_0^{1,2}(\Omega)$.)

$$\int_{\partial\Omega}\left(F(u)x + (x\cdot\nabla u)\nabla u - \frac{1}{2}|\nabla u|^2 x\right)\cdot\nu d\sigma = \int_\Omega\left(NF(u) - \frac{N-2}{2}|\nabla u|^2\right)dx.$$

It is then easy to conclude as in the proof of theorem (2.1.1).

□

Immediately,

Corollary 2.1.4. *Let $f \in C^1(\mathbb{R},\mathbb{R})$ be such that $f(0) = 0$ and let $u \in H^2_{loc}(\mathbb{R}^N)$ be a solution of*

$$\begin{cases}-\Delta u = f(u),\\ \qquad u \in \mathcal{D}^{1,2}(\mathbb{R}^N),\end{cases}$$

such that $F(u) \in L^1(\mathbb{R}^N)$. Then u satisfies

$$\frac{N-2}{2}\int_{\mathbb{R}^N}|\nabla u|^2 = N\int_{\mathbb{R}^N}F(u).$$

Part 3: Pohozaev identity for harmonic map

At the end of this section, motivated by applications, we further verify Pohozaev identity corresponding to the harmonic map.

Theorem 2.1.5. *Suppose that $u : B_1 \to N$ is a smooth harmonic map. For any $t < 0$, in terms of the cylinder coordinates (t, θ), we have*

$$\int_{S^1} |\partial_t u|^2 - |\partial_\theta u|^2 \, d\theta = 0 \tag{2.1.5}$$

and

$$\int_{S^1} \partial_t u \cdot \partial_\theta u \, d\theta = 0. \tag{2.1.6}$$

Proof. We prove (13) only. The harmonic map equation implies that Δu is perpendicular to $\partial_\theta u$, hence

$$\int_{S^1} \Delta u \cdot \partial_\theta u \, d\theta = 0.$$

Using the trivial identities

$$\int_{S^1} \partial_\theta \left(|\partial_\theta u|^2\right) d\theta = \int_{S^1} \partial_\theta \left(|\partial_t u|^2\right) d\theta = 0,$$

we obtain

$$\partial_t \int_{S^1} \partial_t u \cdot \partial_\theta u \, d\theta = 0.$$

To finish the proof of (13), we integrate the above equation over $[t, \infty)$ and notice that

$$\lim_{t \to \infty} \int_{S^1} \partial_t u \cdot \partial_\theta u \, d\theta = 0$$

since u is smooth at the origin.

□

2 Several Conformally Invariant Variational Problems

In this section we concern on several conformally invariant variational problems, each has its corresponding geometric background.

Prescribed Mean Curvature Equation

Let $B_1 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 1\}$. For $u \in C^1(\overline{B}_1, \mathbb{R}^3)$, we consider the functional

$$\mathcal{F}(u) = \frac{1}{2} \int_{B_1} |Du|^2 + \int_{B_1} u^* \omega, \tag{2.2.1}$$

where $\omega \in C^1(\mathbb{R}^3, \Lambda_2(\mathbb{R}^3))$ is a given 2-form. We compute the first variation formula of this functional.

Theorem 2.2.1. *Let $\mathcal{F}(u)$ be the functional in (2.2.1), $d\omega = H dV_{\mathbb{R}^3}$. Then for any $u \in C^2(\overline{B}_1, \mathbb{R}^3)$ and $\varphi \in C^2_c(B_1, \mathbb{R}^3)$, we have*

$$\frac{d}{d\varepsilon}\mathcal{F}(u+\varepsilon\varphi)\Big|_{\varepsilon=0} = \int_{B_1} \langle Du, D\varphi\rangle + \int_{B_1} H \circ u \langle u_x \wedge u_y, \varphi\rangle. \tag{2.2.2}$$

Proof. Consider the affine homotopy map F:

$$F : D \times [0,\varepsilon] \to \mathbb{R}^3, \quad F(z,t) = u(z) + t\varphi(z).$$

Applying Stokes' theorem on $D \times [0,\varepsilon]$, we get

$$\int_{B_1 \times [0,\varepsilon]} F^*(H dV_{\mathbb{R}^3}) = \int_{B_1} (u+\varepsilon\varphi)^*\omega - \int_{B_1} u^*\omega + \int_{\partial B_1 \times [0,\varepsilon]} F^*\omega. \tag{2.2.3}$$

Claim: The last term on the right-hand side of (2.2.3) is zero. In fact, we assume $\omega = \omega_{ij} dy^i \wedge dy^j$, then

$$\begin{aligned}
F^*\omega(r,t) &= \omega_{ij}(F) dF^i \wedge dF^j \\
&= \omega_{ij}(F)(\frac{\partial F^i}{\partial x}dx + \frac{\partial F^i}{\partial y}dy + \frac{\partial F^i}{\partial t}dt) \wedge (\frac{\partial F^j}{\partial x}dx + \frac{\partial F^j}{\partial y}dy + \frac{\partial F^j}{\partial t}dt) \\
&= \omega_{ij}(F)(\frac{\partial F^i}{\partial x}dx + \frac{\partial F^i}{\partial y}dy) \wedge (\frac{\partial F^j}{\partial x}dx + \frac{\partial F^j}{\partial y}dy) \\
&= \omega_{ij}(F)(\frac{\partial F^i}{\partial x}\frac{\partial F^j}{\partial y} - \frac{\partial F^i}{\partial y}\frac{\partial F^j}{\partial x}) dx \wedge dy.
\end{aligned}$$

The last step is because $\frac{\partial}{\partial t}F(z,t) = \varphi(z) = 0$ on $\partial B_1 \times [0,\varepsilon]$. By (2.2.3), differentiating at $\varepsilon = 0$ yields

$$\begin{aligned}
\frac{\partial}{\partial\varepsilon}\int_{B_1}(u+\varepsilon\varphi)^*\omega\Big|_{\varepsilon=0} &= \frac{\partial}{\partial\varepsilon}\int_0^\varepsilon \int_D H(F(x,y,t)) \det DF(x,y,t) dxdydt\Big|_{\varepsilon=0} \\
&= \int_{B_1} H(u(x,y)) \det DF(x,y,0) dxdy \\
&= \int_{B_1} H(u(x,y)) \det(u_x, u_y, \varphi) dxdy \\
&= \int_{B_1} H(u(x,y)) \langle u_x \wedge u_y, \varphi\rangle dxdy.
\end{aligned}$$

$\square$

We can easily see that the positive limit point of F is the solution to the following elliptic equation system:

$$\Delta u = (H \circ u) u_x \wedge u_y \quad \text{in} \quad D. \tag{2.1}$$

We can verify that u satisfies the conditions of Theorem 1.4, thus establishing a conformal relation.

To derive the prescribed mean curvature equation from the first variation formula, we assume that u has a positive property. However, this cannot be obtained from the existence theorem, so the key to solving this problem lies in how to obtain these solutions with positive properties. Below, we first study special cases: constant mean curvature equations, and then general cases.

Constant Mean Curvature Equation [CMC]

Assume the mean curvature H is a constant. We study the analytical properties of the solution $u \in W^{1,2}(D^2, \mathbb{R}^3)$ of the following equation:

$$\Delta u - 2H \partial_x u \times \partial_y u = 0. \tag{2.2}$$

We have an intuition that the Jacobian structure in (2.5) ensures that there are no excessive complications in the research process. We first prove that the Palais-Smale sequence is weakly convergent.

Let F_n be a sequence in $H^{-1}(D^2, \mathbb{R}^3)$ that converges strongly to 0 in $H^{-1}(D^2)$. Let u_n be a uniformly bounded sequence in $W^{1,2}$ that satisfies the equation

$$\Delta u_n - 2H \partial_x u_n \times \partial_y u_n = F_n \to 0 \quad \text{strongly in} \quad H^{-1}(D^2).$$

We write

$$(\partial_x u_n \times \partial_y u_n)^i = \partial_x u_n^{i+1} \partial_y u_n^{i-1} - \partial_x u_n^{i-1} \partial_y u_n^{i+1} = \partial_x (u_n^{i+1} \partial_y u_n^{i-1}) - \partial_y (u_n^{i+1} \partial_x u_n^{i-1}).$$

Since u_n is uniformly bounded in $W^{1,2}$, there exists a subsequence $u_{n'}$ that converges weakly to u_∞. By the Rellich-Kondrachov compactness theorem, u_n converges strongly in L^2. Specifically, we can take the limit of the second term:

$$u_n^{i+1} \partial_y u_n^{i-1} \to u_\infty^{i+1} \partial_y u_\infty^{i-1} \quad \text{in} \quad D'(D^2)$$

and

$$u_n^{i+1}\partial_x u_n^{i-1} \to u_\infty^{i+1}\partial_x u_\infty^{i-1} \quad \text{in} \quad D'(D^2).$$

Combining (2.6), we get that u_∞ is a solution to the CMC equation (2.5). Since the right-hand side of the CMC equation is quadratic, the Calderon-Zygmund theory fails. To obtain the positivity of the solution to the CMC equation, we need more preparatory work. More precisely, we need to use Wente's theorem [14].

Theorem 2.2.2. *For any $a, b \in W^{1,2}(B^2)$, let $\phi \in W_0^{1,p}(D^2)$ be the unique solution to the following equation, where $1 \le p < 2$:*

$$\begin{cases} -\Delta\phi = \partial_x a \partial_y b - \partial_x b \partial_y a & \text{in } D^2 \\ \phi = 0 & \text{on } \partial D^2 \end{cases}.$$

Then $\phi \in C^0 \cap W^{1,2}(D^2)$ and satisfies

$$\|\phi\|_{L^\infty(D^2)} + \|\nabla\phi\|_{L^2(D^2)} \le C_0 \|\nabla a\|_{L^2(D^2)} \|\nabla b\|_{L^2(D^2)}.$$

Here, C_0 depends on a and b.

Proof First, for the sake of simplifying calculations, we assume that a and b are smooth functions. For general a and b, we only need a simple approximation. The continuity of ϕ can be obtained through the approximation of smooth functions.

We can use integration by parts and the Cauchy-Schwarz inequality to estimate

$$\begin{aligned} \int_{D^2} |\nabla\phi|^2 = -\int_{D^2} \phi\Delta\phi &\le \|\phi\|_\infty \|\partial_x a \partial_y b - \partial_x b \partial_y a\|_1 \\ &\le 2\|\phi\|_\infty \|\nabla a\|_2 \|\nabla b\|_2. \end{aligned}$$

Thus, if $\phi \in L^\infty$, then naturally $\phi \in W^{1,2}$.

Based on the above analysis, we first consider the functions $\tilde{a}, \tilde{b} \in C_0^\infty(\mathbb{C})$. Since $C_0^\infty(\mathbb{C})$ is dense in $W^{1,2}(\mathbb{C})$, to obtain the estimate (2.8), we let

$$\tilde{\phi} := \frac{1}{2\pi} \ln \frac{1}{r} \otimes [\partial_x \tilde{a} \partial_y \tilde{b} - \partial_x \tilde{b} \partial_y \tilde{a}].$$

By polar coordinate transformation, we can get

$$|\tilde{\phi}(0)| \le C_0 \|\nabla \tilde{a}\|_{L^2(\mathbb{C})} \|\nabla \tilde{b}\|_{L^2(\mathbb{C})}.$$

In fact,

$$\tilde{\phi}(0) = -\frac{1}{2\pi}\int_{\mathbb{R}^2} \ln r \left(\partial_x \tilde{a} \partial_y \tilde{b} - \partial_x \tilde{b} \partial_y \tilde{a}\right)$$

$$= -\frac{1}{2\pi}\int_0^{2\pi}\int_0^{+\infty} \ln r \left[\frac{\partial}{\partial r}\left(\tilde{a}\frac{\partial \tilde{b}}{\partial \theta}\right) - \frac{\partial}{\partial \theta}\left(\tilde{a}\frac{\partial \tilde{b}}{\partial r}\right)\right] dr d\theta$$

3 Hilbert's XIXth Problem

Hilbert's XIXth problem (1900): Consider any local minimizer of energy functionals of the form

$$\mathcal{E}(w) := \int_\Omega L(\nabla w)\, dx, \tag{2.3}$$

where $L : \mathbb{R}^n \to \mathbb{R}$ is smooth and uniformly convex, and $\Omega \subset \mathbb{R}^n$. Is it true that all local minimizers to this type of problems are smooth?

Chapter 3 Fundamental Issues in Harmonic Map

1 The ε-Regularity Theorem

We first introduce the classical ε-Regularity Theorem, as proven by Sacks and Uhlenbeck in their paper, "*The Existence of Minimal Immersions of 2-Spheres*".

Approximate the integral functional $E(u) = \int_M |\nabla u|^2 d\mu$, whose critical points are harmonic maps, by a slightly different integral. For convenience, we choose a measure $d\mu$ on M so that the area of M equals 1. Let

$$E_\alpha(s) = \int_M (1 + \sum_{i=1}^{k} \langle ds^i(x), ds^i(x)\rangle)^\alpha d\mu = \int_M (1 + |ds(x)|^2)^\alpha d\mu.$$

For $\alpha = 1$, $E_1(s) = 1 + E(s)$ has harmonic maps as critical points. For $\alpha > 1$, E_α is well-behaved. The Sobolev space of maps

$$L_1^{2\alpha}(M, N) = \{s \in L_1^{2\alpha}(M, R^k) \colon s(x) \in N\} \subset C^0(M, N)$$

is a C^2 separable Banach manifold for $\alpha > 1$. This, and the following theorem, plus several other basic smoothness theorems can be found in Palais [P3].

Theorem 3.1.1. *Supppose there exists $\varepsilon > 0$ and $\alpha_0 > 1$ such that if $u \colon D \to N$ is a smooth critical point of $E_\alpha \overset{set}{=} \int_M (1 + |\nabla u|^2)^\alpha$, $E(u) < \varepsilon$ and $1 \le \alpha < \alpha_0$, then there is an estimate uniform in $1 \le \alpha < \alpha_0$,*

$$\|\nabla u\|_{D',1,p} \le C(p, D')\|\nabla u\|_{0,2}$$

for $D' \Subset D$ any smaller disk.

Lemma 3.1.2. *If $u_t = \Delta u + A(u)(\nabla u, \nabla u)$ in B_r, where $r \leq 1$. If $\sup_{P_{2r}} |\nabla u| \leq 2$, then*

$$\sup_{P_r} |\nabla u|^2 \leq C \frac{1}{r^4} \int_{-4r^2}^{0} \int_{B_{2r}} |\nabla u|^2 \, dx \, dt$$

Proposition 3.1.3. *Let $u_t = \Delta u + A(u)(\nabla u, \nabla u)$ in P_1, and $\sup_{-1 \leq t \leq 0} \int_{B_1} |\nabla u|^2 \, dx \leq \varepsilon_1$, then*

$$\sup_{P_{\frac{1}{2}}} |\nabla u|^2 \leq C \int_{-1}^{0} \int_{B_1} |\nabla u|^2 \, dx \, dt.$$

Theorem 3.1.4. *Suppose $P_1 = B_1(0) \times [-1, 0]$, u is harmonic map heat flow on P_1, and $\Lambda \overset{set}{=} \int_{-1}^{0} \int_{B_1(0)} |\nabla u|^2 dx dt \leq \varepsilon_0$, then*

1) $\sup_{B_{\frac{2}{3}}} |\nabla u|(\cdot, 0) \leq C$,

2) $\sup_{B_{\frac{1}{3}}} |\nabla u|(\cdot, 0) \leq C\Lambda$

holds.

Proof. By Lemma (3.1), 1) implies 2), hence it suffices to prove 1).

$$\sup_{-\frac{1}{2} \leq t \leq 0} \int_{B_{\frac{1}{2}}} |\nabla u|^2 \, dx \leq \int_{B_1} |\nabla u|^2 \, dx + C \frac{\varepsilon_0}{4}$$

$$\int_{B_1} |\nabla u|^2(t) \, dx = \int_{B} \varphi |\nabla u|^2 \, d(tz) = \int_{B} \varphi |\nabla u|^2(z_t) + \int_{t_1}^{0} \int_{B} 2\varphi_t u_z \nabla u \, dz \, dt$$

$$\leq \int_{B_1} |\nabla u|^2(z_t) - \int_{t_1}^{0} \int_{B} 2\varphi |\nabla u_t|^2 \, dz \, dt + \int_{t_1}^{0} \int_{B_1} 8\varphi |u_z|^2 \, dz \, dt + \frac{C}{8} \int_{t_1}^{0} \int_{B_1} |\nabla u|^2 \frac{|\nabla \varphi|^2}{\varphi} \, dz \, dt$$

$$\leq \int_{B_1} |\nabla u|^2(z_t) + \int_{t_1}^{0} \int_{B} C |\nabla u|^2 \frac{1}{(R+t)^2} \, dz \, dt$$

we get a contradiction.

□

$$\partial_t u - \Delta_{\mathcal{M}} u + \Gamma_{\mathcal{N}}(u)(\nabla u, \nabla u)_{\mathcal{M}} = 0,$$

$$u|_{t=0} = u_0,$$

Theorem 3.1.5. *There exists a constant* $\varepsilon_0 > 0$ *depending only on* $\mathcal{N}$ *and* m *such that for any regular solution* $u\colon \mathbb{R}^m \times [-4R_0^2, 0] \to \mathcal{N}$ *of with* $E(u(t)) \leq E_0 < \infty$*, uniformly in* t*, the following is true:*

If for some $R \in]0, R_0[$ *there holds*

$$\Psi(R) := \Psi(R; u) = \int_{T_R} |\nabla u|^2 G \, dx \, dt < \varepsilon_0, \tag{3.1}$$

then

$$\sup_{P_{\delta R}} |\nabla u|^2 \leq c(\delta R)^{-2} \tag{3.2}$$

with constants $\delta > 0$ *depending on* $\mathcal{N}, m, E_0,$ *and* $\inf\{R, 1\}$*, and* c *depending on* $\mathcal{N}$ *and* m*, only.*

Theorem 3.1.6. *For any* $R_0 > 0$ *and* E_0 *there exists a constant* $\varepsilon_0 > 0$ *depending on* R_0*,* E_0*,* $\mathcal{N}$*, and* m *such that for any regular solution* $u\colon \mathbb{R}^m \times [-4R_0^2, 0] \to \mathcal{N}$ *of with* $E(u(t)) \leq E_0 < \infty$*, the following is true:*

If for some $R \in]0, R_0[$ *there holds*

$$\Psi(R; u) = \int_{T_R} |\nabla u|^2 G \, dx \, dt < \varepsilon_0, \tag{3.3}$$

then

$$\sup_{P_{R/4}} |\nabla u|^2 \leq CR^{-2} \tag{3.4}$$

with a constant C *depending on* $\mathcal{N}$ *and* m *only.*

2 Removable Singularity Theorem

3 Energy Identity

4 The Formation and Properties of Bubble Trees

The bubble tree construction above associates to each bubble point x a sequence of neck maps $\phi_n : [0, T_n] \times S^1 \to M$ whose images are thin tubes in M. Using these, we can associate to x two numerical quantities:

$$\begin{cases} \text{energy loss} & \tau(x) & = \limsup_n E(\phi_{n,x}) \\ \text{neck length} & \nu(x) & = \limsup_n \sup_{p,q \in [0,T_n]\times S^1} \mathrm{dist}(\phi_n(p), \phi_n(q)). \end{cases}$$

In this section we will prove that both these numbers vanish.

Lemma 3.4.1. *At each bubble point x we have $\tau(x) = 0$ and $\nu(x) = 0$.*

Lemma 2.1 completes our analysis of the bubble tree, and, with the results of §1, immediately gives our main convergence theorem.

Theorem 3.4.2 (Bubble Tree Convergence for Harmonic Maps)**.** *Let $\{f_n\} : \Sigma \to M$ be a sequence of harmonic maps from a fixed Riemann surface (Σ, h) to a compact Riemannian manifold (M, g) with $E(f_n) \le E_0$. Then there is a subsequence $\{f_n\}$ and a bubble tower domain T so that the renormalized maps*

$$\{f_{n,I}\} : T \to M$$

converge in $L^{1,2} \cap C^0$ to a smooth harmonic bubble tree map $\{f_I\} : T \to M$. Moreover,

(a) (No energy loss) $E(f_n)$ converges to $\sum E(f_I)$, and

(b) (Zero distance bubbling) At each bubble point x_J (at any level in the tree), the images of the base map f_I and the bubble map f_J meet at $f_I(x_J) = f_J(p^-)$.

Consequently the image of the limit $\{f_I\} : T \to M$ is connected, and the images of the original maps $f_n : \Sigma \to M$ converge pointwise to this image $\{f_I\}$.

Chapter 4 Topics in Harmonic Map and PDE

1 Higer-order Neck Analysis of Harmonic Maps and Application

Theorem 4.1.1. *Suppose u_i is a sequence of harmonic maps satisfying (i–iii). There exist uniformly bounded coefficients p_i, q_i, a_i, b_i, c_i and d_i such that for any $\alpha \in (0,1)$, $t \in [\log \lambda_i/\delta, \log \delta]$,*

$$u_i(t,\theta) = p_i + q_i t + a_i e^t \cos\theta + b_i e^t \sin\theta + c_i \lambda_i e^{-t} \cos\theta + d_i \lambda_i e^{-t} \sin\theta + O(\eta^{1+\alpha}), \tag{2}$$

where $O(\eta^{1+\alpha})$ is some function uniformly bounded by $C\eta^{1+\alpha}$.

Moreover,

$$\big||q_i|^2 - 2\lambda_i(a_i \cdot c_i + b_i \cdot d_i)\big| \leq C\lambda_i^{\alpha/2+1}, \tag{3}$$

and

$$|a_i \cdot d_i - b_i \cdot c_i| \leq C\lambda_i^{\alpha/2}. \tag{4}$$

2 Lojasiewicz Inequality and Its Applications in Harmonic Map Theory

This work presents a comprehensive treatment of the Lojasiewicz inequality and its fundamental role in the analysis of harmonic map heat flows. We establish the theoretical foundation of this inequality for real analytic functions, provide key proof techniques, and demonstrate its critical application in establishing convergence results for harmonic map flows. The unified presentation highlights the deep connection between real analyticity and the long-time behavior of geometric evolution equations.

The Lojasiewicz Inequality

The Lojasiewicz inequality, introduced by Stanislaw Lojasiewicz in the 1960s, is a profound result in real analytic geometry that quantifies the behavior of analytic functions near critical points.

Definition 4.2.1 (Real analytic function)**.** *A function $f : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}$ is real analytic if for every point in the open set Ω, it admits a convergent power series expansion.*

Theorem 4.2.2 (Lojasiewicz inequality)**.** *Let $f : \Omega \to \mathbb{R}$ be real analytic with a critical point at $a \in \Omega$ ($\nabla f(a) = 0$). There exist constants $C > 0$, $\theta \in (0, 1)$, and a neighborhood $U \subset \Omega$ of a such that for all $x \in U$:*

$$|\nabla f(x)| \geq C|f(x) - f(a)|^\theta. \tag{4.1}$$

The exponent θ depends on the local geometry of f at a.

Proof. The core proof strategy is presented for polynomial functions, with extension to general analytic cases.

Step 1. Polynomial case

Assume f is polynomial, $a = 0$, $f(0) = 0$, and $\nabla f(0) = 0$. Let $m \geq 2$ be the order of vanishing of f at 0. The Taylor expansion decomposes as:

$$f(x) = P_m(x) + R(x), \quad \nabla f(x) = \nabla P_m(x) + \nabla R(x)$$

where P_m is homogeneous of degree m, and $R(x)$ contains higher-order terms. By homogeneity, there exist $c_1, d_1 > 0$ such that:

$$|P_m(x)| \geq c_1|x|^m,$$
$$|\nabla P_m(x)| \geq d_1|x|^{m-1}.$$

In a neighborhood of 0, the remainder satisfies $|R(x)| \leq K|x|^{m+1}$ and $|\nabla R(x)| \leq K|x|^m$. For sufficiently small $|x|$:

$$|f(x)| \geq \tfrac{c_1}{2}|x|^m,$$
$$|\nabla f(x)| \geq \tfrac{d_1}{2}|x|^{m-1}.$$

Combining these with $\theta = \frac{m-1}{m}$ yields:

$$|\nabla f(x)| \geq C|f(x)|^\theta, \quad C = \tfrac{d_1}{2}\left(\tfrac{2}{c_1}\right)^\theta.$$

Step 2. General analytic case

For general analytic functions, the convergent expansion $f(x) = \sum_{k=m}^\infty P_k(x)$ allows similar estimates. The complete proof requires the curve selection lemma from analytic set theory. □

Geometric Interpretation

- **Exponent** θ: Measures critical point flatness (e.g., $\theta = 1/2$ for $f(x) = x^2$, $\theta = 3/4$ for $f(x) = x^4$)
- **Analyticity necessity**: The inequality may fail for C^∞ non-analytic functions (e.g., $f(x) = e^{-1/x^2}\sin(1/x)$ at $x = 0$)
- **Geometric significance**: Controls the steepness of descent directions near critical points

Applications to Harmonic Map Theory

Let (M, g) be a compact Riemannian manifold with boundary and (N, h) a complete Riemannian manifold.

Definition 4.2.3 (Harmonic maps). *A map $\phi : M \to N$ is harmonic if it is a critical point of the energy functional:*

$$E(\phi) = \frac{1}{2}\int_M |d\phi|^2 dvol_g.$$

The Euler-Lagrange equation is $\tau(\phi) = trace_g \nabla d\phi = 0$, where τ denotes the tension field.

The harmonic map heat flow (gradient flow of E) is given by:

$$\frac{\partial \phi_t}{\partial t} = \tau(\phi_t), \quad \phi|_{t=0} = \phi_0. \tag{4.2}$$

The fundamental question is whether ϕ_t converges to a harmonic map as $t \to \infty$.

Convergence via Lojasiewicz Inequality

When N is analytic (e.g., spheres $\mathbb{S}^n$, projective spaces $\mathbb{RP}^n$, compact Lie groups), the energy functional satisfies a Lojasiewicz inequality near critical points.

Theorem 4.2.4 (Heat flow convergence)**.** *Let ϕ_∞ be a harmonic map where E satisfies the Lojasiewicz inequality:*

$$\|\nabla E(\phi)\|_{L^2} = \|\tau(\phi)\|_{L^2} \geq C|E(\phi) - E(\phi_\infty)|^\theta.$$

If the initial map ϕ_0 is sufficiently close to ϕ_∞, then the heat flow ϕ_t exists globally and converges smoothly to a harmonic map as $t \to \infty$.

Proof. Along the heat flow, $\frac{d}{dt}E(\phi_t) = -\|\tau(\phi_t)\|_{L^2}^2$. Substituting the Lojasiewicz inequality yields:

$$\frac{d}{dt}\Big[E(\phi_t) - E(\phi_\infty)\Big] \leq -C^2\Big|E(\phi_t) - E(\phi_\infty)\Big|^{2\theta}.$$

Setting $\Psi(t) = E(\phi_t) - E(\phi_\infty)$, we solve the differential inequality:

$$\begin{aligned} \theta = \tfrac{1}{2} &\Rightarrow \Psi(t) \leq \Psi(0)e^{-Ct} \quad \text{(exponential convergence)} \\ \theta < \tfrac{1}{2} &\Rightarrow \Psi(t) \leq Kt^{-\frac{1}{1-2\theta}} \quad \text{(polynomial convergence)} \end{aligned}$$

Compactness arguments and elliptic estimates establish smooth convergence $\phi_t \to \phi_\infty$. □

Simon's Fundamental Theorem

L. Simon's landmark result systematizes this convergence:

Theorem 4.2.5 (Simon's convergence theorem (1983))**.** *If N is analytic and the initial energy $E(\phi_0)$ is close to some harmonic map energy, the heat flow ϕ_t converges smoothly to a harmonic map.*

Mechanism and Geometric Insight

The Lojasiewicz inequality resolves critical difficulties in geometric flows:

- **Prevents stagnation**: Eliminates non-convergent behaviors where $\|\tau(\phi_t)\| \to 0$ but $E(\phi_t)$ doesn't approach a critical value
- **Quantifies convergence**: The exponent θ determines convergence rates
- **Handles singularities**: Controls energy decay during bubbling phenomena
- **Ensures smoothness**: Combines with elliptic regularity to guarantee smooth limits

Table 4.1: Convergence rates in harmonic map flow

Critical point type	Exponent θ	Convergence rate
Non-degenerate	1/2	Exponential
Degenerate ($m = 4$)	3/4	t^{-3}
General analytic	$\in (0, 1)$	$t^{-\kappa/(1-2\theta)}$

Extensions and Limitations

- **Non-analytic targets**: For C^∞ but non-analytic N, convergence may fail (oscillatory behaviors observed)
- **Singularity analysis**: Controls energy decay rates near singular times
- **Other geometric flows**: Analogous results for mean curvature flow, Yang-Mills flow, etc.
- **Optimal exponents**: θ related to Łojasiewicz exponent in geometric contexts

The Lojasiewicz inequality provides a deep connection between real analyticity and the asymptotic behavior of geometric evolution equations. By establishing a fundamental relationship between gradient norms and functional values near critical points, it resolves convergence issues that plague non-analytic settings. In harmonic map theory, this inequality serves as the cornerstone for understanding the long-time behavior of geometric flows, demonstrating the profound interplay between local analytic geometry and global geometric analysis.

Chapter 5 Some Specific Harmonic maps and PDEs

1 Regularity in Optimal Transportation

Definition 5.1.1 (Optimal Transportation)**.** *Given $\Omega, \Omega^* \subset \mathbb{R}^n$, and two mass densities $\mu = f\,dx$ in Ω and $\nu = g\,dy$ in Ω^*, satisfying*

$$\mu(\Omega) = \nu(\Omega^*).$$

A map $T : \Omega \to \Omega^*$ which minimises the total cost

$$\mathcal{C}(t) = \int_\Omega f(x)c\big(x, t(x)\big)\,dx \quad t \in S$$

(where $c(x, y)$ is the cost function) is called

Here are some Example of cost functions

- $c(x, y) = |x - y|$ (natural cost, Monge's cost)
- $c(x, y) = |x - y|^2$ (quadratic cost)
- $c(x, y) = |x - y|^p$ $(p \neq 0,)$
- $c(x, y) = -\log|x - y|$ (light reflection)
- $c(x, y) = \log x \cdot y$ (Aleksandrov's problem)

The cost function $c(x, y) = |x - y|^2$ is equivalent to $c(x, y) = x \cdot y$.

Kantorovich also introduced the dual functional

$$\mathcal{I}(\varphi, \psi) \overset{set}{=} \int_\Omega \varphi(x)f(x) + \int_{\Omega^*} \psi(y)g(y), \quad (\varphi, \psi) \in K$$

$$K \overset{set}{=} \{(\varphi, \psi) : \varphi(x) + \psi(y) \leq c(x, y)\}.$$

Then the following duality holds:

$$\inf\{\mathcal{C}(s) : s \in S\} = \sup\{\mathcal{I}(\varphi, \psi) : \quad (\varphi, \psi) \in K\}.$$

Here are some advantages of studying the functional $\mathcal{I}$:

- The dual functional is linear,
- The set K is convex,
- $\mathcal{I}$ deals with functions but $\mathcal{C}$ deals with mappings.

Now introduce potential functions. For any pair $(u, v) \in K$, let

$$\begin{aligned} u^*(x) &\stackrel{set}{=} \inf\{c(x,y) - v(y) : \quad y \in \Omega^*\} \\ v^*(y) &\stackrel{set}{=} \inf\{c(x,y) - u(x) : \quad x \in \Omega\}. \end{aligned}$$

Then (u^*, v^*) is uniformly Lipschitz, and

$$\begin{aligned} \mathcal{I}(u^*, v^*) - \mathcal{I}(u, v) &= \int_\Omega \big(u^*(x) - u(x)\big) f(x) + \int_{\Omega^*} \big(v^*(y) - v(y)\big) g(y) \\ &= \end{aligned}$$

We also have $\mathcal{I}(u-a, v+a) = \mathcal{I}(u, v)$ for any constant a. Hence there exists a sequence of Lipschitz maximizing sequences which converges to the maximizer (u, v). The maximizer (u, v) are called potential functions, and

$$\mathcal{I}(u, v) = \sup\{\mathcal{I}(\varphi, \psi) : (\varphi, \psi) \in K\}.$$

The potential functions (u, v) are dual to each other:

$$\begin{aligned} u(x) &= \inf\{c(x,y) - v(y) : y \in \Omega^*\}, \\ v(y) &= \inf\{c(x,y) - u(x) : x \in \Omega\}. \end{aligned}$$

This is the Legendre transform if $c(x, y) = x \cdot y$.

(u, v) are Lipschitz continuous, and

$$\|u, v\|_{\text{Lip}} \le \|c\|_{\text{Lip}}.$$

Therefore the gradient Du exists a.e..

From (*), $\forall x \in \Omega$, $\exists y \in \Omega^*$ such that

$$u(x) = c(x, y) - v(y)$$
$$u(x') \le c(x, y) - v(y) : \forall x' \in \Omega.$$

Differentiating the first formula,

$$Du(x) = D_x c(x, y).$$

(**) is the starting point for the existence and regularity. Application 2: Aleksandrov's problem

Given $f \in C^0(S^n)$, find a convex $M = \{x\rho(x) : x \in S^n\}$ such that the integral Gauss curvature of M at $x\rho(x)$ is equal to $f(x)$.

Theorem. ρ is a solution to Aleksandrov's problem iff $u = \log \rho$ is a maximiser to

$$l(u, v) = \int_\Omega f(x)u(x) + \int_{\Omega^*} g(y)v(y)$$

among functions (u, v) subject to the restriction

$$u(x) + v(y) \le c(x, y)$$

where

$$c(x, y) = \log x \cdot y$$

In the above two examples, the cost functions are

$$c(x, y) = -\log |x - y|$$

$$c(x, y) = \log x \cdot y$$

2 Embedded Hypersurface in $\mathbb{R}^{n+1}$

Let Σ be an n-dimensional differentiable hypersurface in $\mathbb{R}^{n+1}$. We denote by g or $\langle \cdot, \cdot \rangle$ the induced metric on Σ and by A the second fundamental form corresponding to a unit normal field ν. In local coordinates, we write:

$$g = g_{ij}\, dx^i \otimes dx^j$$
$$A = h_{ij}\, dx^i \otimes dx^j \quad \Rightarrow \quad |A|^2 = g^{ik} g^{jl} h_{ij} h_{kl}.$$

The mean curvature H is given by $H = g^{ij}h_{ij}$. (Prop 3.1.4)

The Levi-Civita connection is denoted by ∇.

$$\nabla_\Sigma A = \nabla_i h_{jk}\, dx^i \otimes dx^j \otimes dx^k$$
$$|\nabla_\Sigma A|^2 = g^{ik}g^{jl}g^{pq}\nabla_i h_{jp}\nabla_k h_{lq}.$$

We now derive a differential identity, which will be used in the derivation of the interior gradient estimates.

Lemma 5.2.1. *Let Σ be a C^4-embedded hypersurface in $\mathbb{R}^{n+1}$ and ν be its unit normal field. Then*

$$\Delta_\Sigma \nu + |A|^2\nu + \nabla_\Sigma H = 0 \quad \text{on } \Sigma.$$

Proof. Step 1: $\forall p \in \Sigma$, we introduce Riemann normal coordinates with orthonormal tangent vectors $z_1, \ldots, z_n$ and s.t. $\Gamma^k_{ij} = 0$ at p. Set $D_i = D_{z_i} \to$ directional derivatives, then

$$\begin{aligned} D_iD_j &= \nabla_i\nabla_j \quad \text{at } p, \\ \nabla_i z_j &= \nabla_{z_i} z_j + h_{ij}\nu = h_{ij}\nu, \\ -\langle D_i\nu, z_j\rangle &= h_{ij} \quad (\Rightarrow A(x,y) = -\langle D_x\nu, y\rangle), \\ &\Rightarrow D_i\nu = -h_{ij}z_j, \\ \text{and} \quad \Delta_\Sigma\nu &= g^{ij}\nabla_i\nabla_j\nu = D_iD_i\nu. \end{aligned}$$

The Codazzi equations yield at p

$$D_k h_{ij} = D_j h_{ik}.$$

Step 2: We now compute at p:

$$\begin{aligned} \Delta_\Sigma\nu = D_iD_i\nu &= -D_i(h_{ij}z_j) = -(D_ih_{ij})z_j - h_{ij}D_iz_j \\ &= -(D_ih_{ij})z_j - h_{ij}^2\nu \\ &= -\nabla_\Sigma H - |A|^2\nu \end{aligned}$$

where $H = h^{ii}$ is the mean curvature function. $\square$

In the rest of this section, we derive a differential inequality which plays an essential role in the discussion of curvature estimates.

Lemma 5.2.2 (Simons Identity). *Let Σ be a C^4-embedded hypersurface in $\mathbb{R}^{n+1}$. Then*

$$\Delta_\Sigma h_{ij} = \nabla_i \nabla_j H + H g^{kl} h_{ik} h_{lj} - |A|^2 h_{ij}.$$

Proof. Step 1: By the Codazzi equation, we have

$$\begin{aligned} \nabla_k h_{ij} &= \nabla_i h_{kj} \\ \Rightarrow \nabla_k \nabla_i h_{ij} &= \nabla_k \nabla_i h_{ij}. \end{aligned}$$

By exchanging the order of differentiation, we get

$$\begin{aligned} \nabla_k \nabla_i h_{ij} &= \nabla_i \nabla_k h_{ij} - R^p_{kil} h_{pj} - R^p_{kij} h_{lp} \\ &= \nabla_i \nabla_k h_{ij} - g^{pq} R_{kilp} h_{pj} - g^{pq} R_{kijq} h_{lp} \\ \nabla_i \nabla_j h_{kc} &\to \text{Codazzi equation} \\ \Rightarrow \Delta_\Sigma h_{ij} &= g^{kl} \nabla_k \nabla_l h_{ij} = g^{kl} \nabla_i \nabla_j h_{kl} - g^{kl} g^{pq} R_{kilp} h_{pj} - g^{kl} g^{pq} R_{kijq} h_{lp} \\ \nabla_i \nabla_j H &= \nabla_i \nabla_j (g^{kl} h_{kl}) = \nabla_i (g^{kl} \nabla_j h_{kl}) = g^{kl} \nabla_i \nabla_j h_{kl}. \end{aligned}$$

Step 2: The Gauss equation yields

$$\begin{aligned} & g^{kl} g^{pq} R_{kilq} h_{pj} + g^{kl} g^{pq} R_{kijq} h_{lp} \\ = g^{kl} g^{pq} (h_{kq} h_{ic} - h_{kc} h_{iq}) h_{pj} &+ g^{kl} g^{pq} (h_{kq} h_{ij} - h_{kj} h_{iq}) h_{lp} \\ 1 - 4 &= g^{kl} g^{pq} (h_{kq} h_{ic} h_{pj} - h_{kj} h_{iq} h_{pl}) = 0 \\ &= g^{kl} g^{pq} h_{kq} h_{ic} h_{pj} - g^{pq} g^{kl} h_{pj} h_{ic} h_{kq} = 0 \\ 2 = -g^{kl} g^{pq} h_{kc} h_{iq} h_{pj} &= -H g^{pq} h_{iq} h_{pj} = -H g^{kl} h_{ik} h_{lj} \\ 3 &= g^{kl} g^{pq} h_{kq} h_{ij} h_{lp} = |A|^2 h_{ij} \end{aligned}$$

□

Lemma 3: Let Σ be a C^4-embedded hypersurface in $\mathbb{R}^{n+1}$. Then

$$\tfrac{1}{2} \Delta_\Sigma |A|^2 = |\nabla A|^2 + h^{ij} \nabla_i \nabla_j H + H h^i_i h^j_j h^k_k - |A|^4$$

where $\quad h_{ij} = g_{ik} h_{kj} \quad$ and $\quad h^{ij} = g^{ik} g^{jl} h_{kl}.$

Proof: Step 1:

$$
\begin{aligned}
|A|^2 &= g^{ik}g^{jl}h_{ij}h_{kl} \qquad (17)\\
\Rightarrow \tfrac{1}{2}\Delta_\Sigma|A|^2 &= \tfrac{1}{2}\Delta_\Sigma(g^{ik}g^{jl}h_{ij}h_{kl})\\
&= \tfrac{1}{2}g^{pq}\nabla_p\nabla_q(g^{ik}g^{jl}h_{ij}h_{kl})\\
&= \tfrac{1}{2}g^{pq}g^{ik}g^{jl}\nabla_p h_{kl}\nabla_q h_{ij} + g^{ik}g^{jl}h_{ij}\nabla_p\nabla_q h_{kl}\\
&= g^{ik}g^{jl}h_{kl}\Delta_\Sigma h_{ij} + g^{ik}g^{jl}g^{pq}\nabla_p h_{kl}\nabla_q h_{ij}.
\end{aligned}
$$

Step 2: By Lemma 2 and (17),

$$
\begin{aligned}
g^{ik}g^{jl}h_{kl}\Delta_\Sigma h_{ij} &= g^{ik}g^{jl}h_{kl}(\nabla_i\nabla_j H + Hg^{pq}h_{ip}h_{qj} - |A|^2h_{ij})\\
&= g^{ik}g^{jl}h_{kl}\nabla_i\nabla_j H + g^{ik}g^{jl}g^{pq}h_{kl}h_{ip}h_{qj}H - |A|^4.
\end{aligned}
$$

As a consequence, we prove an estimate of the second fundamental form for minimal hypersurfaces, which plays an essential role in the discussion of integral curvature estimates.

Lemma 5.2.3. *Let Σ be a C^4-minimal hypersurface in $\mathbb{R}^{n+1}$. Then*

$$
\frac{1}{2}\Delta_\Sigma|A|^2 = \left(1+\frac{2}{n}\right)|\nabla_\Sigma A|^2 - |A|^4.
$$

Proof. Step 1: Since $H=0$, then

$$
\frac{1}{2}\Delta_\Sigma|A|^2 = |\nabla_\Sigma A|^2 - |A|^4.
$$

We need to estimate $|\nabla_\Sigma A|^2$. ($\nabla_k h_{ij} = h_{ij,k}$)

By the Codazzi equation, $h_{jk,i} = h_{ik,j}$.

$\forall p\in\Sigma$, introduce Riemann normal coordinates near p s.t. $z_1,\dots,z_n$ are orthonormal, $\Gamma_{ij}^k = 0$ at p.

At p, $|A|^2 = \frac{1}{n}h_i^i$, $|\nabla_\Sigma A|^2 = \frac{1}{n^2k}h_{i,j}^2h_{j,k}^2$.

h_{ij} is diagonal.

Step 2: Claim: $|\nabla_\Sigma|A||^2 \le \frac{n}{|A|^2}h_{ii,k}^2$

To verify this, we consider the case $|A(p)|^2 \to 0$. Then, at p,

$$
\begin{aligned}
|\nabla_\Sigma |A||^2(p) &= \left|\nabla_\Sigma \sqrt{g^{ik}g^{jl}h_{ij}h_{kl}}\right|^2(p) \\
&= \left|\frac{1}{2}\frac{\nabla_\Sigma(g^{ik}g^{jl}h_{ij}h_{kl})}{\sqrt{g^{ik}g^{jl}h_{ij}h_{kl}}}\right|^2(p) \\
&= \frac{1}{4|A|^2}\nabla_k(h_{ij}h_{ij,k})^2 \\
&= \frac{1}{4|A|^2}\nabla_k(h_{ii}h_{ii,k})^2.
\end{aligned}
$$

By Cauchy inequality, $|\nabla_\Sigma|A||^2 \le \frac{n}{|A|^2}h_{ii,k}^2$.

Step 3: Claim: $|\nabla_\Sigma A|^2 \ge (1+\frac{2}{n})|\nabla_\Sigma|A||^2$.

$$
\begin{aligned}
|\nabla_\Sigma A|^2 - |\nabla_\Sigma|A||^2 &\ge \sum_{ijk} h_{ij,k}^2 - \sum_{ijk} h_{ii,k}^2 \\
&= \sum_{ijk} h_{ij,k}^2 \\
&= \sum_{ijk} h_{ij,i}^2 + \sum_{ijk} h_{ij,j}^2.
\end{aligned}
$$

$h_{ij,i} = \nabla_i h_{ij} = \nabla_i h_{ji} = \nabla_j h_{ii}$

$\sum_{ijk} h_{ij,i}^2 = \sum_{i\ne j} h_{ij,i}^2$

$\Rightarrow |\nabla_\Sigma A|^2 - |\nabla_\Sigma|A||^2 \ge 2\sum_{i\ne j} h_{ij,i}^2$

$$
\begin{aligned}
0 = H = g^{ij}h_{ij} &\Rightarrow g^{ij}h_{ij,k} = 0 \\
&\Rightarrow g^{ij}h_{ij,k}(p) = h_{ii,k}(p) = 0 \\
&\Rightarrow h_{iii} = -\frac{n}{n-1}h_{jj,i} \\
&\Rightarrow |\nabla_\Sigma|A||^2 \le \sum_{i,k} h_{iik}^2 = \sum_i\sum_k h_{iik}^2 + \sum_i h_{iii}^2 \\
&= \sum_i\sum_k h_{iik}^2 + \sum_i\left(\frac{n}{n-1}h_{jj,i}\right)^2 \\
&\le \sum_i\sum_k h_{iik}^2 + (n-1)\sum_i\sum_j h_{jj,i}^2 \\
&= n\sum_i\sum_j h_{jj,i}^2 \\
&\Rightarrow |\nabla_\Sigma A|^2 - |\nabla_\Sigma|A||^2 \ge \frac{2}{n}|\nabla_\Sigma|A||^2.
\end{aligned}
$$

By Cauchy inequality.

□